

THE FACTORY COORDINATOR FOR AGENTIC DELIVERY

CONVERGE

Coordinate intent, decomposition, task authority, execution, and settlement – **without duplicating authority**.

An end-to-end technical book: nine passes, two phases, one human barrier.

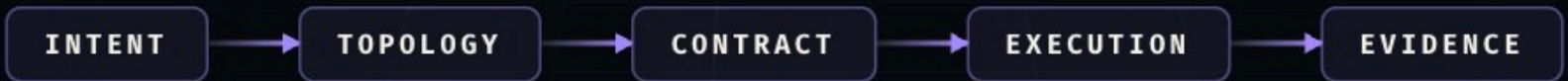


Settlement Fold

PROCESS IDENTITY • THE FOLD THAT SETTLES

cvg 0.2.0 (task-spec 3.8.0)

THE CONVERGE SYSTEM – CANONICAL EBOOK • CONVERGE 0.2.0

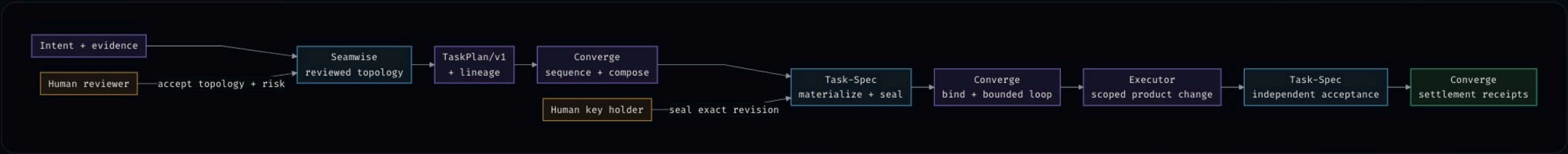


The referee, not the player.

It sequences authority without duplicating it, binds one authorized task to a bounded runtime, and settles only after inspectable evidence.

“Converge is the factory coordinator and assurance layer around independent decomposition and task-contract engines... *It is the referee, not the player.*”

— brf-converge · §1 North star ♦ *the one exact mental model this book must leave behind*



BEAT 1 Unease many agents can act, but authority becomes ambiguous	BEAT 2 Separation topology, authorization, execution, acceptance, observation are different jobs	BEAT 3 Descent intent becomes progressively more falsifiable across nine passes	BEAT 4 Barrier a human accepts the adversarially reviewed topology before build authority begins	BEAT 5 Bounded motion one sealed revision receives one runtime contract and one finite loop	BEAT 6 Settlement green evals, path policy, independent acceptance, and receipts agree	BEAT 7 Honesty release, current main, working tree, and roadmap stay distinct
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A README, a green test, and a release receipt are not interchangeable evidence.

Every claim in this book carries a source label – the rank says how much weight that label can bear.

RANK	SOURCE	SURFACE	USE IN THIS BOOK
1	A Authority	OPERATING.md · AGENTS.md · docs/concepts/authority.md	Ownership, non-authority, merge and factory/everyday boundaries.
2	C Contract	contracts/ · schemas · VERSION · CLI matrix	Machine-visible shapes, commands, tokens, versions, mutation flags.
3	I Implemented	bin/ · skills/*/scripts/ · apps/cockpit/	Executable behavior at the audited committed HEAD.
4	T Tested	tests/ · skill suites · direct 2026-08-26 runs	Claims directly discriminated by observed checks.
5	E Evidence	evidence/releases/ · immutable receipts and snapshots	Retained execution proof; invalidated corridors remain invalid.
6	R Release	v0.2.0 · release workflow · archived assets	Frozen release claims only; do not import later-main behavior.
7	D Documentary	README.md · docs/ · CHANGELOG.md · existing decks	Explanation and reported hosted evidence; reconcile against code/tests.
8	W Working tree	16 modified + one untracked guided-chat file	W CHECKOUT ONLY – audited but uncommitted behavior.
9	F Future	explicit exclusions · Manager/fleet references	Roadmap or absent behavior – never render as a current capability.

508 paths audited. The subsystem gates are green. The aggregate gate is red.

The audit ran the current checkout against the pinned engines – the counts below are **observed**, not reported.

TRACKED PATHS · SOURCE AUDIT

AREA	PATHS	WHAT WAS INSPECTED
skills/	220	11 packaged skills, 55 scripts, 108 skill-test files, prompts, templates, fixtures
apps/	102	Cockpit client/server, ACP Ask path, schemas, tests, build
evidence/	72	release, invalidated, environment, receipt, acceptance, final-status records
tests/	22	aggregate CLI, compose, install, clean room, loop, package, docs, layout
assets/	21	Settlement Fold brand and process assets
docs/	18	authority, trust, descent, bind/loop, recovery, CLI and knowledge map
scripts/	11	bootstrap, docs, evidence, release, demo, asset validation
bin/	7	referee CLI, composition, lane, snapshot, agent-context, UI helper
.github/	7	CI, release, dependency and ownership configuration
contracts/	6	CLI matrix and versioned JSON Schemas
total tracked	508	baseline 58b1ddb = origin/main

v0.2.0 tag 3de9f0b is 42 commits behind the audited HEAD · checkout expands the CLI matrix 57 → 60 forms

W CHECKOUT ONLY

DIRECT VALIDATION · OBSERVED 2026-08-26

SURFACE	DIRECT RESULT	HONEST INTERPRETATION
Runtime contract	48/48	Pass 7 policy, binding, freshness, topology, adapters, settlement
Register	145/145	Pass 6 parity, idempotency, DAG, receipt, identity, projection
Loop kernel	57/57	brakes, budgets, isolation, tier 2, path + external-write policy
Passes 0–4	39+24+38+33+23	structural pass gates discriminate their named fixtures
Conductor checkout	44/44	guided mode + pre/post routing — uncommitted checkout only
Clean-room path	21/21	copy install → init → seal → bind → loop → accept → done
Compose	14/14	review cannot be skipped; tamper and engine drift fail closed
JSON envelope	25/25	position independence, exit preservation, mutation truth
JSON matrix	60 forms / 239 calls	checkout-only 60-form machine surface is serializable
Docs	DOCS=READY files=22 commands=60	checkout documentation internally aligned
Package	PACKAGE=READY files=244 skills=11	package manifests internally aligned
Cockpit	40 server + 62 client	read-only bridge, bounded Ask, snapshots; production build green
Retained evidence	20 files	named live evidence validates; not a production-duration claim
Archived assets	6/6 on v0.1.0	historical binaries recoverable from their owning tag
Composed demo	LOCAL_SETTLED · ACCEPTED=1 · DEMO_COMPOSED=READY	deterministic local path, default external-write denial

LOAD-BEARING FAILURE · AGGREGATE GATE

Aggregate **make check FAILS** — `tests/test-repo-layout.sh` rejects tracked `AGENTS.md` + `OPERATING.md` as undeclared top-level files. The subsystems above passed; **release-ready may NOT be claimed**. Safe move: name the subsystem gates, remediate the layout contract through an explicit repository decision, then rerun the full aggregate.

Twelve claims this book may make. Eleven it must refuse.

Claim discipline is part of the product – every page that follows is bound by this policy (condensed from §2; the brief rows stay authoritative).

MAY MAKE • GREEN CLAIMS

12 claims

- ✓ A **thin factory coordinator** around independently versioned Seamwise and Task-Spec binaries.
- ✓ Each decision has **one authority**; duplicate capability is tolerated, duplicate authority is not.
- ✓ **Nine passes, two phases, one barrier**; Capture optional, Register opt-in.
- ✓ Materialization never authorizes dispatch — the receipt records `dispatch_authorized: false`.
- ✓ `taskspec gate --stamp` is the **only** engine action that seals a leaf for delegation.
- ✓ Bind adds a repository fence that may be **stricter than the Task-Spec write scope**.
- ✓ The loop chooses no work — **one named issue** under finite iteration, time, token, path budgets.
- ✓ Default isolation is a **Git worktree**; settled work survives on a task branch.
- ✓ **Eight distinct terminal outcomes**: LOCAL_SETTLED, SETTLED, NO_OP, BLOCKED, STALLED, EXHAUSTED, CANCELLED, ERROR.
- ✓ **Task-Spec performs independent acceptance** of the exact handoff and revision; Converge does not reimplement it.
- ✓ **Cockpit and Ask cannot authorize or settle** — observation and interpretation only.
- ✓ Immutable v0.2.0 release and current main are **different evidence corridors**.

MUST REFUSE • RED CLAIMS

11 refusals

- ✗ “Everyday portfolio coordinator” — `OPERATING.md` assigns that plane to **WorkHelm**.
- ✗ Converge **owns Seamwise or Task-Spec internals** (decomposition, authorization, acceptance).
- ✗ A reviewed topology, materialized task, green narration, or composition receipt **authorizes execution**.
- ✗ HMAC **proves identity, hides the key, or sandboxes** — it is tamper evidence under a shared key.
- ✗ Bind **prevents every out-of-scope write** — some adapters only detect after the fact.
- ✗ Tier-2 holdouts are **secret from the worker** — the brief still points the worker at the Task-Spec.
- ✗ `LOCAL_SETTLED` means a PR opened, remote state changed, production healthy, or a merge.
- ✗ Cockpit, Ask, `cvg next`, or a tracker projection is a **second source of canonical state**.
- ✗ Manager fleet, autonomous fan-out, live-tracker or production reliability **ships in v0.2.0**.
- ✗ Checkout-only guided chat and the 60-form CLI are **committed, released, or hosted**.
- ✗ Current `make check` passes — **the layout gate is red**.

Every color is a claim about authority.

Violet coordinates, cyan decomposes, green proves, red refuses — and **gold is spent only on human authority**.

SEMANTIC ROLE	COLOR	USE
Factory Black	near-black	stage, repository floor, operational seriousness
Human Gold	#F3B64C	topology acceptance, barrier, explicit risk decisions — human authority only
Seam Cyan	#68C7FF	Seamwise topology, TaskPlan, lineage, decomposition
Contract Purple	#A78BFA	Converge sequencing, binding, composition, machine contracts
Accepted Green	#3DDC97	verified gates, acceptance, current receipt, safe settlement
Refused Red	#FF6B66	stale review, tamper, scope escape, refutation, unsupported claim
Observer White	#CBD3DA	Cockpit, read-only projections — never authority gold

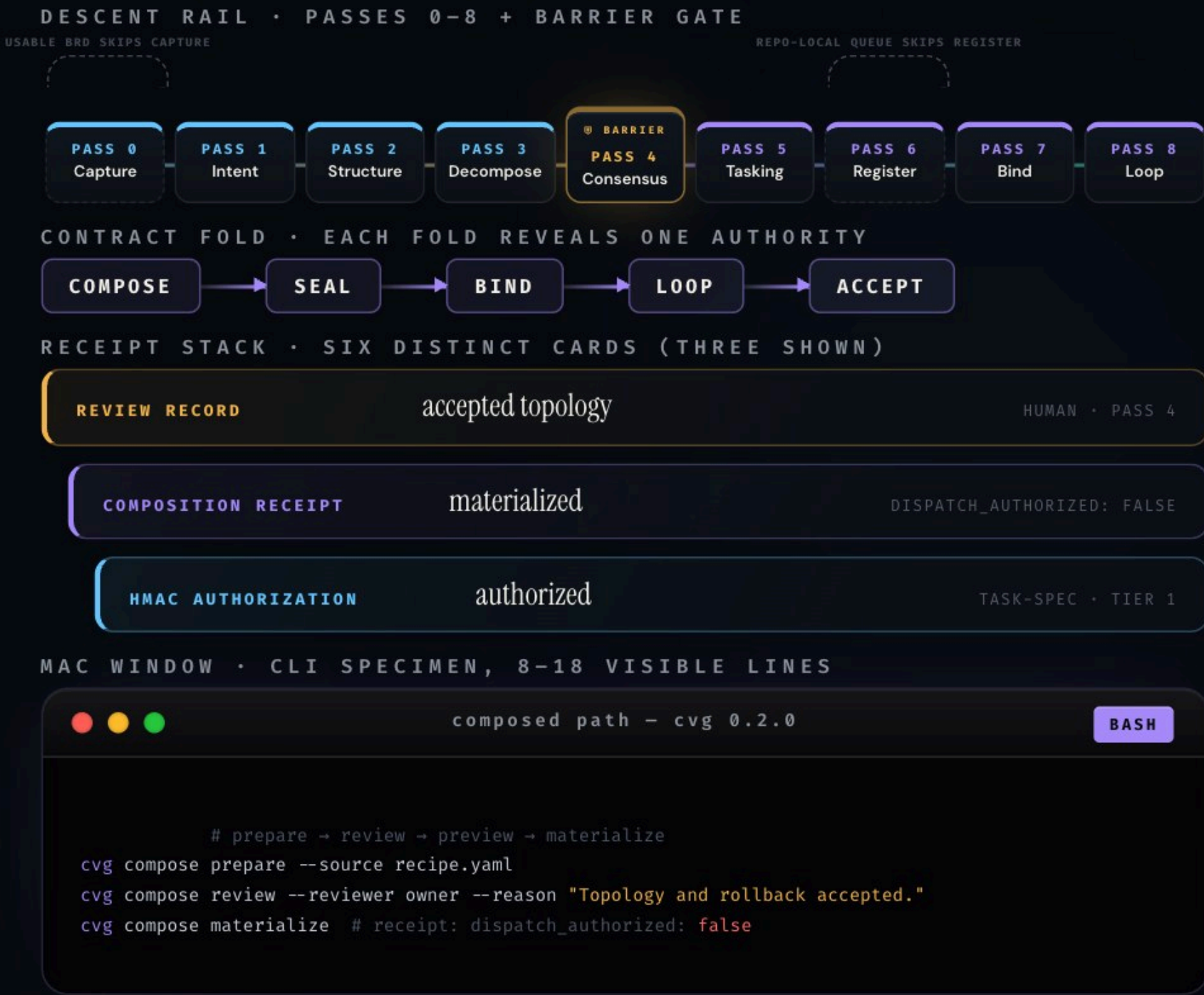
CONNECTOR RULES

- solid + arrowhead** — authoritative route
- dotted** — optional / opt-in route or non-authoritative observation
- NEVER

Cockpit or a model response **never draws a write arrow** into canonical state

DENSITY GUARDRAILS

- One authority decision, one pass, or one trust boundary **per page**.
- One primary visual + at most one compact callout; **≤6 table rows** on narrative pages.
- Every green claim names a **gate, receipt, or test token**; every red claim states **refusal + safe next action**.



Seven acts, one master transition: intent earns settlement.

Each act answers one question and makes one emotional move — the whole book walks a single chain.

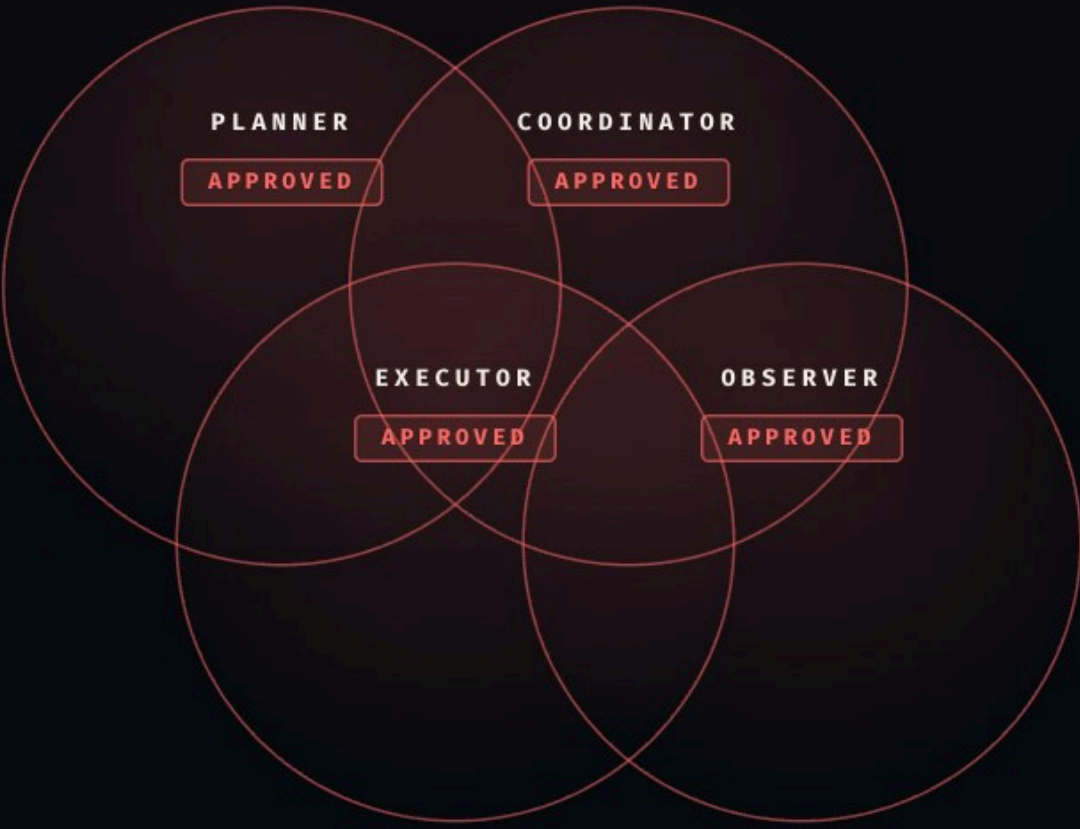


ACT	SLIDES	QUESTION ANSWERED	EMOTIONAL MOVE
I — One decision, one owner ebook pp. 08–14	01–08	What is Converge, and why not another agent framework?	ambiguity → authority
II — The human descent ebook pp. 15–24	09–22	How does raw intent become accepted topology?	uncertainty → agreement
III — Compose and Tasking ebook pp. 25–35	23–36	How does reviewed topology become unsigned, then authorized work?	agreement → explicit authority
IV — Bind the runtime ebook pp. 36–42	37–46	How is one sealed revision constrained before execution?	authority → bounded motion
V — The loop and settlement ebook pp. 43–54	47–60	How does execution stop honestly and settle with proof?	motion → evidence
VI — Operating surfaces ebook pp. 55–63	61–69	How do agents, humans, CLI, and Cockpit observe and steer safely?	evidence → operational control
VII — Product and proof ebook pp. 64–70	70–76	What is shipped, current, checkout-only, proven, and still open?	control → honest confidence

Agentic delivery fails when authority becomes ambient.

When every layer can decompose, authorize, execute, and declare done, **no receipt means what it says.**

FOUR LAYERS · ONE VERB · EVERYONE APPROVES



THE RULE · I

One decision

Decompose, authorize, execute, accept — each verb in the chain is decided **exactly once**, in exactly one place.

THE RULE · II

One owner

Exactly **one named authority** owns each decision. Every other layer executes, observes, or waits.

REFUSAL · 01

model narration ≠ acceptance

A green-sounding story signs nothing. Safe move: wait for the **Task-Spec acceptance record**.

REFUSAL · 02

tracker state ≠ authority

A projection observes canonical state. Safe move: read the **receipt chain**, never the tracker.

REFUSAL · 03

materialization ≠ dispatch

The receipt records `dispatch_authorized: false`. Safe move: **seal the exact revision** first.

REFUSAL · 04

observation ≠ control

Cockpit reads; it cannot authorize or settle. Safe move: **route authority through its owner**.

It is a referee, not the player.

Converge **sequences independent engines and verifies boundaries**. It does not absorb their authority.

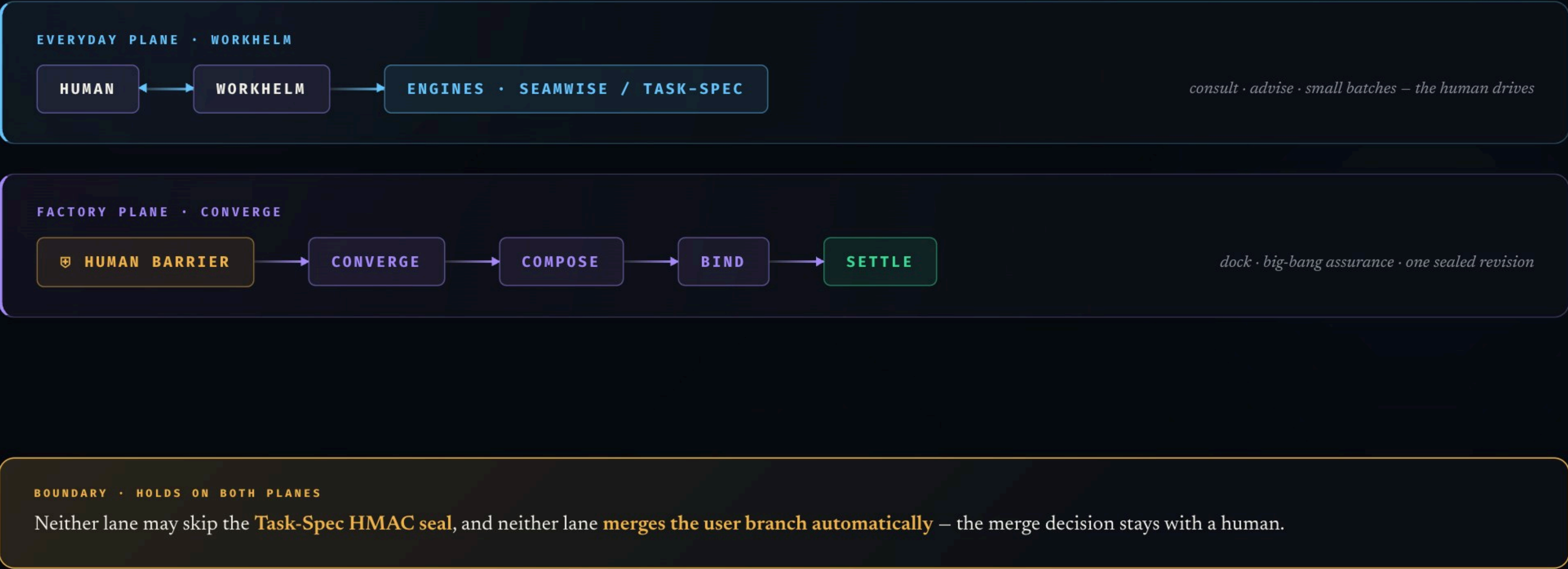


CONVERGE OWNS	CONVERGE DOES NOT OWN
cross-engine sequence	seam discovery
composition receipts	Task-Spec rendering
runtime binding	HMAC authorization
bounded loop + settlement	independent acceptance internals

Thinness is the feature: a coordinator that absorbs its engines becomes the ambient authority it exists to prevent.

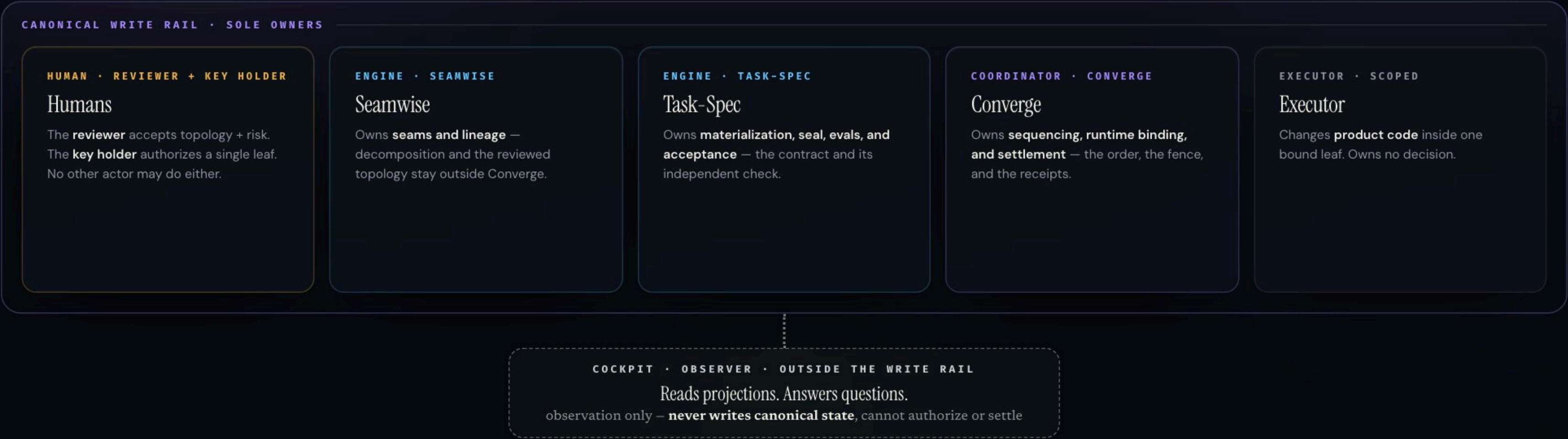
Everyday work and factory work are different planes.

WorkHelm is the everyday plane. Converge is the dock, factory, and big-bang assurance plane.



Duplicate capability is tolerable. Duplicate authority is not.

Five actors may write into the chain — each owns exactly one verb. **Cockpit only ever reads.**



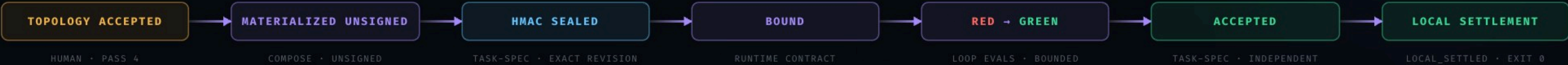
Follow one feature, not nine abstract passes.

A reviewed health endpoint becomes **one unsigned leaf**, **one sealed revision**, **one bounded attempt**, and **one accepted result**.

T-20260815-health-status

THE STEEL THREAD · ONE FEATURE TRACED THROUGH EVERY LATER ACT

STATE OF THE SPECIMEN · AFTER EACH STAGE



The same card reappears across Acts II–VII – the same leaf, the same sealed revision, the same bounded attempt, the same accepted result, at every zoom level.

Six receipts, six different claims.

A review record, composition receipt, HMAC seal, runtime contract, execution receipt, and acceptance record **answer different questions**.

REVIEW RECORD	accepted topology <i>who accepted the topology and the risk?</i>	HUMAN · PASS 4
COMPOSITION RECEIPT	materialized <i>what was composed? still unsigned – dispatch_authorized: false</i>	CONVERGE · COMPOSE
HMAC SEAL	authorized <i>which exact revision may be delegated? tamper evidence under a shared key</i>	TASK-SPEC · GATE --STAMP
RUNTIME CONTRACT	bounded <i>under what fence, budgets, and isolation?</i>	CONVERGE · BIND
EXECUTION RECEIPT	executed <i>what actually ran, and how did it stop?</i>	CONVERGE · LOOP
ACCEPTANCE RECORD	accepted <i>may this exact handoff and revision settle?</i>	TASK-SPEC · INDEPENDENT

REFUSAL · DO NOT FLATTEN

composition receipt ≠ authorization – only the HMAC seal authorizes a leaf.
execution receipt ≠ acceptance – only Task-Spec accepts the handoff. Safe move: cite each receipt for its own verb.

WHERE THE STACK LIVES

composition schema `contracts/converge-composition-receipt-v1.schema.json`

live evidence `evidence/releases/v0.2.0/live-codex/`

Nine passes, two phases, one barrier.

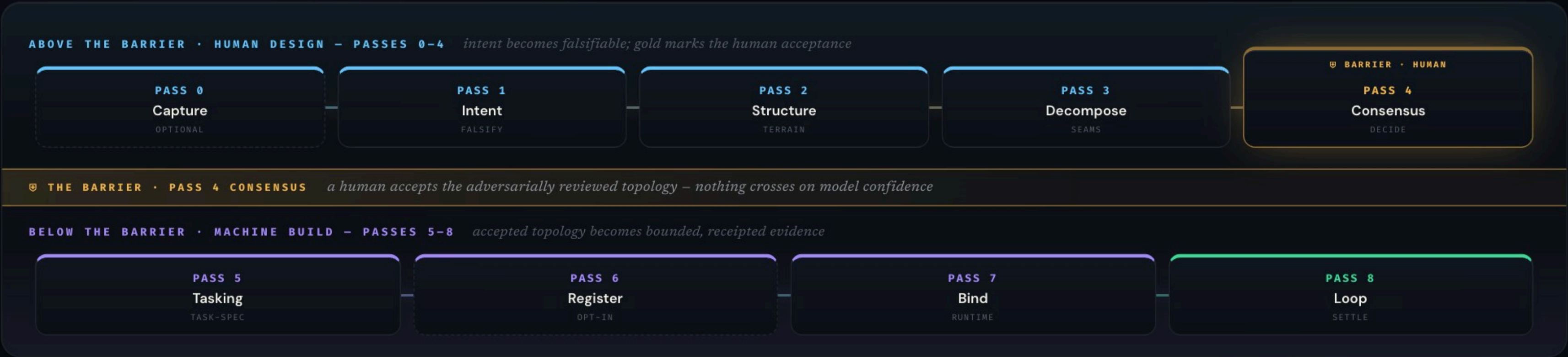
Design above the barrier. Machine build below it. Capture is optional; Register is opt-in.



The first half is not automation theater. It is the work of making intent falsifiable.

Design above the barrier. Build below it.

Passes 0–4 make intent precise. **Passes 5–8 turn accepted intent into bounded evidence.**

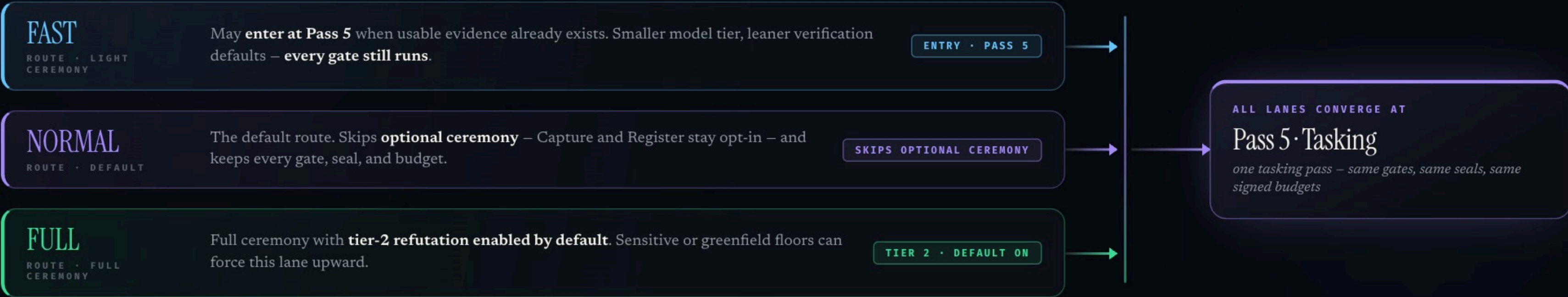


BOUNDARY • MACHINE-LED ≠ SELF-AUTHORIZED

Below the barrier the machines lead the work – but the **HMAC seal still belongs to Task-Spec and an explicit key holder**. No pass below this line authorizes itself.

FAST, NORMAL, FULL route the work. They do not erase proof.

cvg lane changes ceremony, model tier, and verification defaults – **within hard floors**.

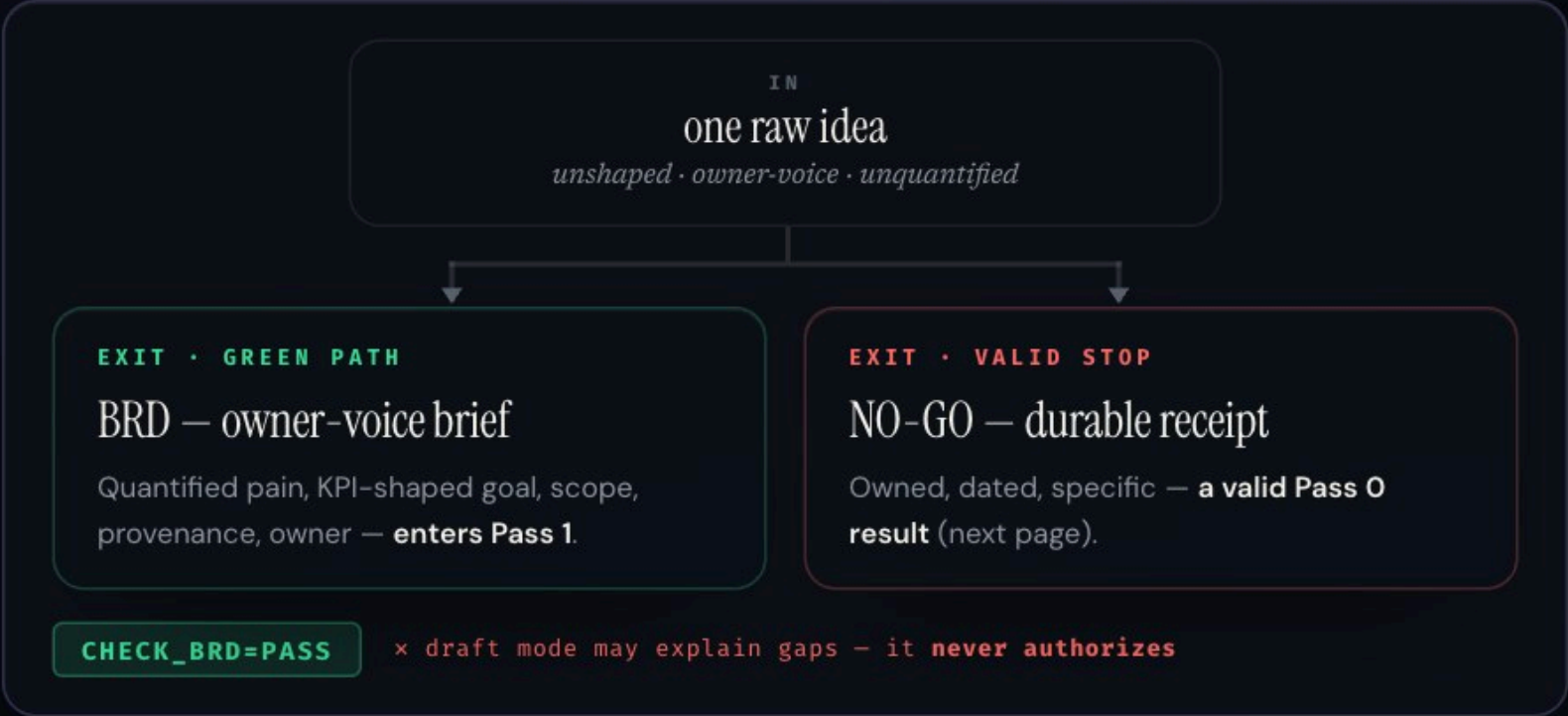


HARD FLOOR · NO LANE CROSSES A lane may tighten ceremony, model tier, and verification defaults – **it cannot widen a signed budget or waive a gate**. Floors push lanes **stronger**, never weaker.

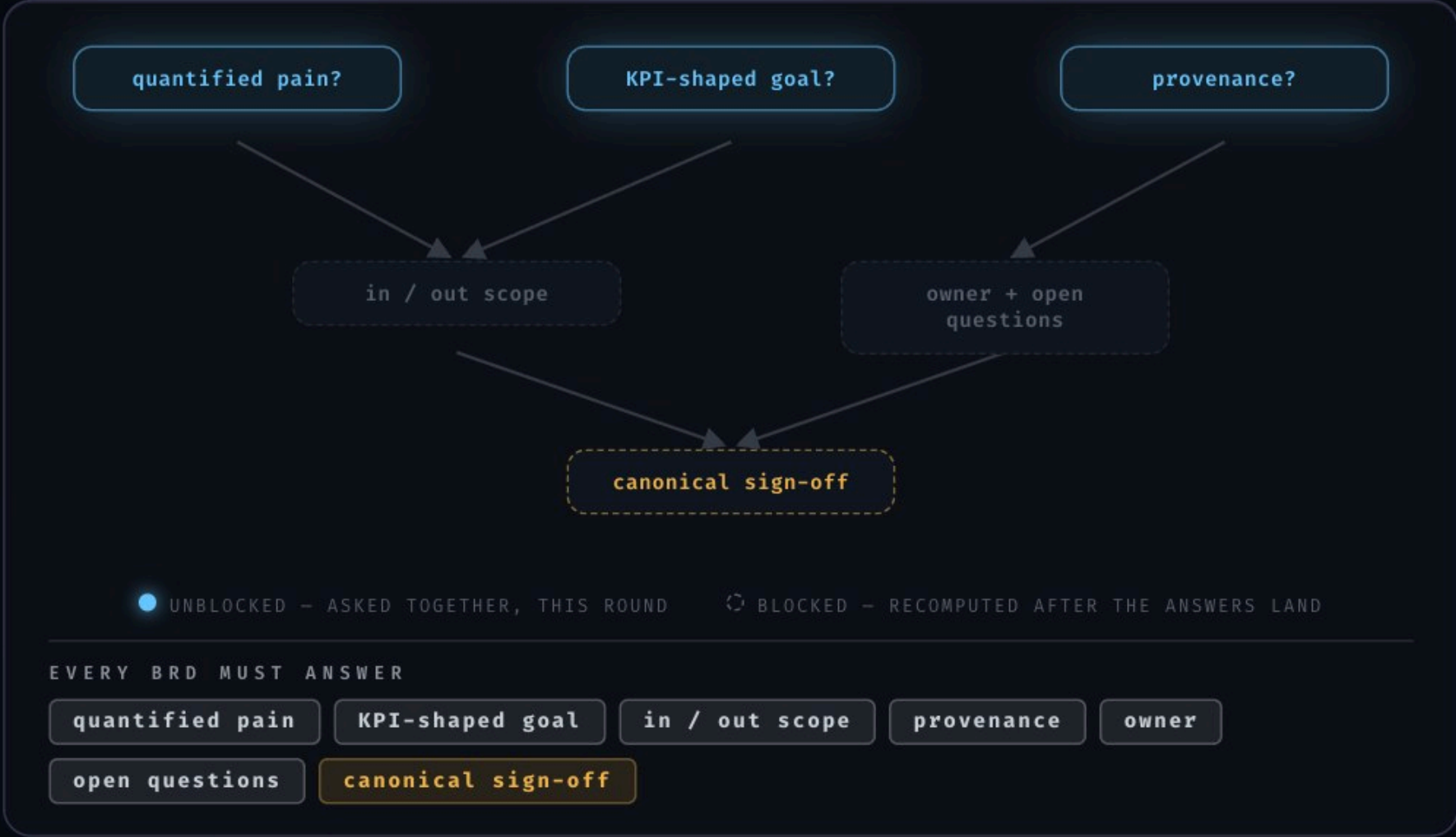
Capture is an interview, not a summarizer.

Raw idea in. **Owner-voice BRD** or **durable no-go out**. Frontier rounds ask every currently unblocked question together – wait once, recompute.

PASS 0 • THE FORK – EXACTLY ONE EXIT

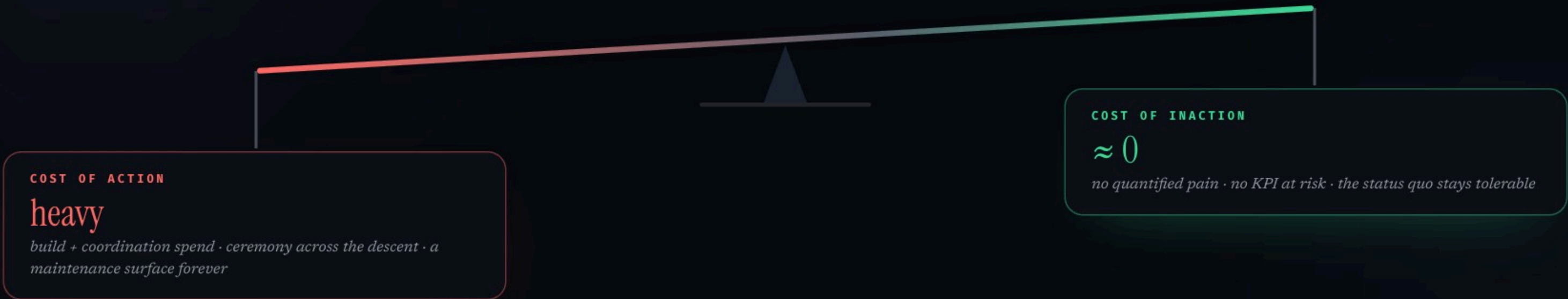


FRONTIER ROUNDS • INTERVIEW CONTROL



If inaction has no meaningful cost, stop manufacturing a project.

The do-nothing test is a kill switch — a **durable no-go** is a successful **Pass 0 outcome**, not a failed project.



NO-GO RECEIPT

exits the descent
durable · owned · dated · specific

PASS 0 • VALID RESULT

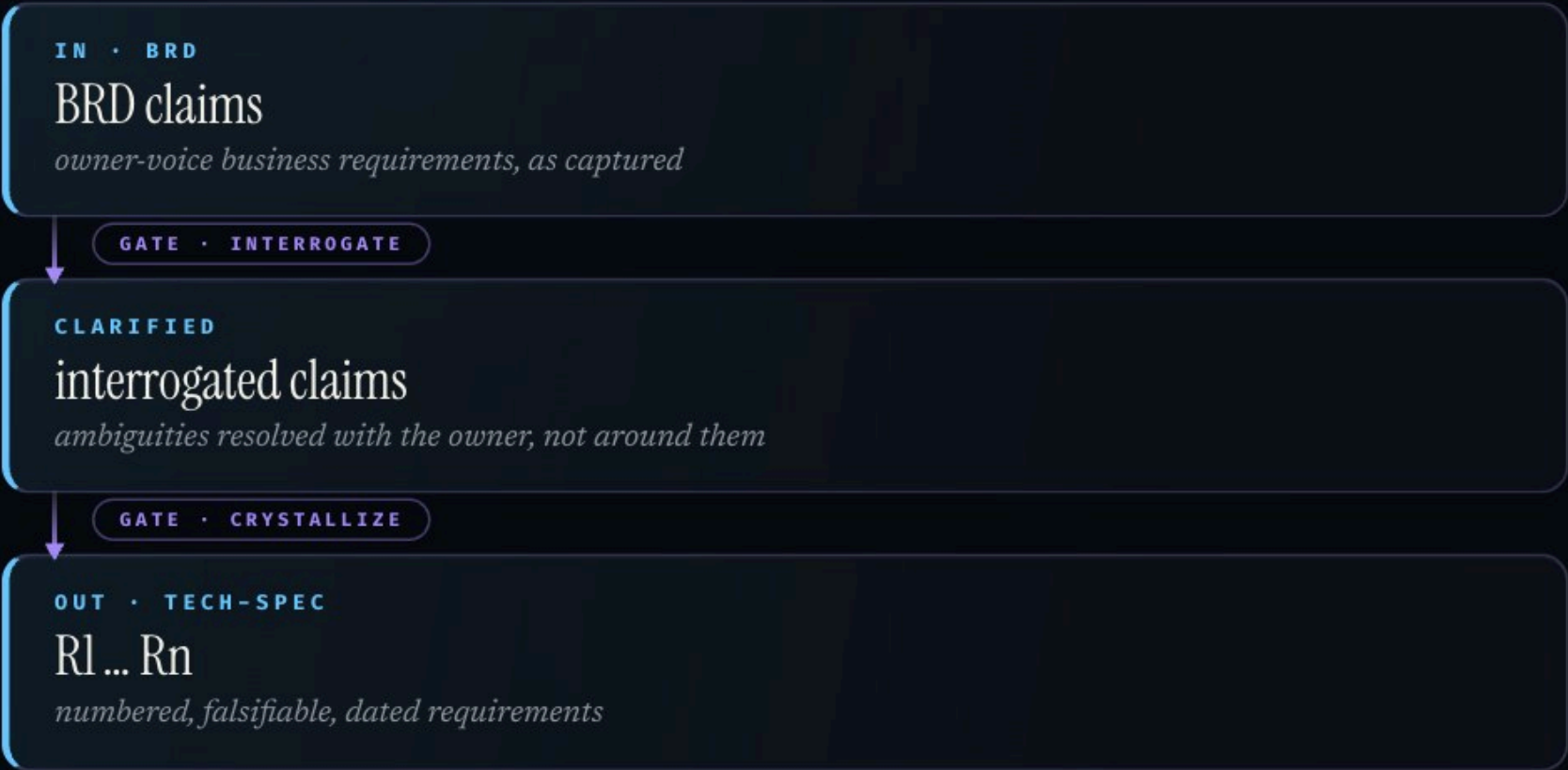
BOUNDARY • NOT MODEL RELUCTANCE

A no-go must be **durable, owned, dated, and specific** — a named human’s decision record, not a model’s hesitation. Safe next action: file the receipt; revisit only when the cost of inaction changes.

Requirements say what must be true. They do not choose architecture.

Client problem in. **Falsifiable tech-spec out** – observable behavior, constraints, and blockers; never infrastructure preferences disguised as requirements.

BRD → REQUIREMENTS • TWO GATES



CHECK_TECH_SPEC=PASS

× unsigned • pending • open blockers • invalid dates • ambiguous inputs – fail closed

ALTITUDE • WHAT, NOT HOW



Record what is true about the ground. Leave building to the next pass.

Tech-spec plus real repository in. **Grounding ADRs** and a **shared context glossary** out.

REPOSITORY TERRAIN • PINNED REQUIREMENTS

TERRAIN • THE AUDITED REPOSITORY, NOT THE IMAGINED ONE

A

finite loop budgets

terrain: **skills/task-loop/** – iterations, wall time, tokens are already bounded surfaces

B

one JSON result contract

terrain: **bin/cvg • contracts/** – every public form speaks one machine envelope

C

read-only observation

terrain: **apps/cockpit/** – snapshot projections exist; no write path back

CHECK_ADR=OK

× final mode rejects proposed ADRs • dangling supersedes • empty evidence • missing context

VERB WALL • ALTITUDE CONTROL

GREEN • RECORD

✓ record what is true about the ground

✓ name terrain: surfaces, constraints, ownership

✓ cite evidence; mark supersession honestly

RED • DRIFT VERBS

build

create

implement

refactor

migrate

belong to the next pass — drift-verb fixtures prove the gate rejects them.

ADR STRUCTURE • THE GROUNDING RECORD

CONTEXT → DECISION → EVIDENCE → CONSEQUENCES

STATUS → SUPERSESSION → GLOSSARY

A seam separates. A swimlane owns. A leg names an observable capability state.

ADRs in. One typed swimlane tree with capability legs out – decomposition stays at plan altitude.

MONOLITH → SEAMS → SWIMLANES → LEGS



GRAMMAR • NOUN 1

seam

separates responsibility — the boundary a plan may cut along.

GRAMMAR • NOUN 2

swimlane

owns the seam — exactly one owner; custody is never shared.

GRAMMAR • NOUN 3

capability leg

names an observable capability state — not yet an atomic Task-Spec.

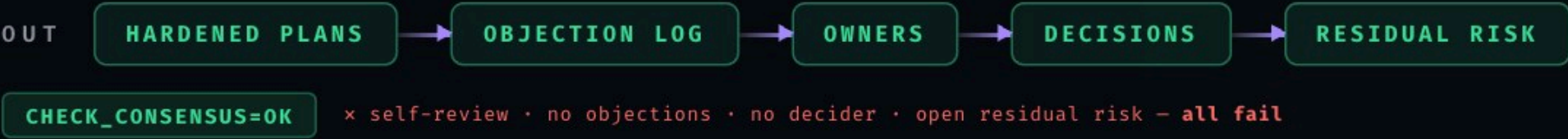
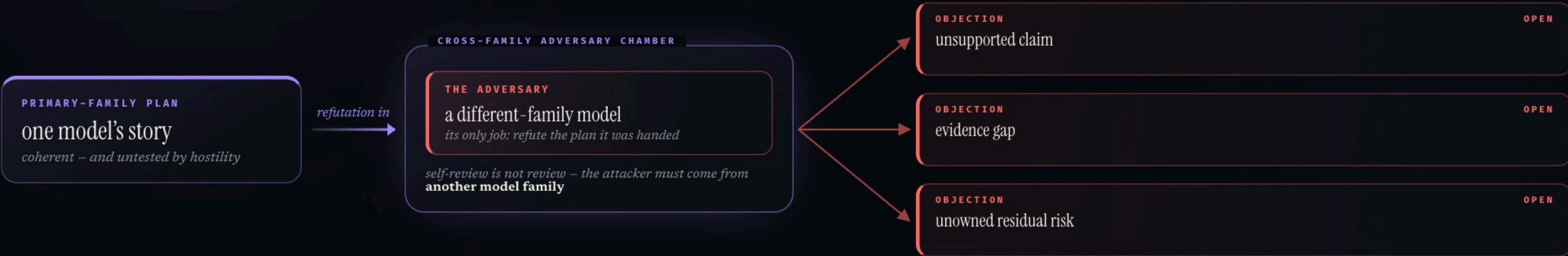
CHECK_PLAN=OK

× no Mermaid • no non-goals • no legs • orphan legs • missing proof — all fail

real task IDs at this altitude = altitude drift

A different-family model tries to break the plans.

Refutation in. **Hardened plans, objection log, owners, decisions, and residual risk out.**



dispatching the adversary **does not close the pass**

An objection exists until a human decides it.

The adversary proposes. A named human decides **FIX** or **ACCEPT** – and owns the residual risk.

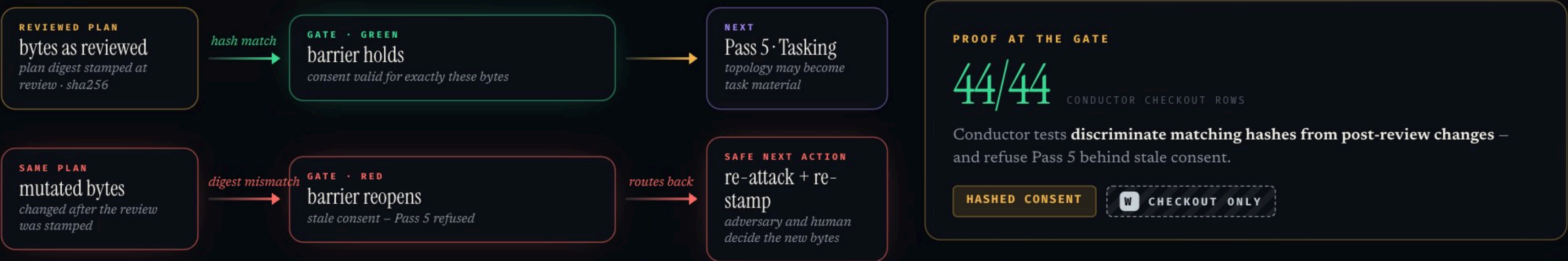


BOUNDARY · SCRUTINY ≠ CONSENSUS
A fresh review log with unresolved findings is evidence of **scrutiny, not consensus**. Safe move: resolve every finding through **DECIDED_BY**, then re-run the consensus gate.

If a reviewed plan changes, the barrier reopens.

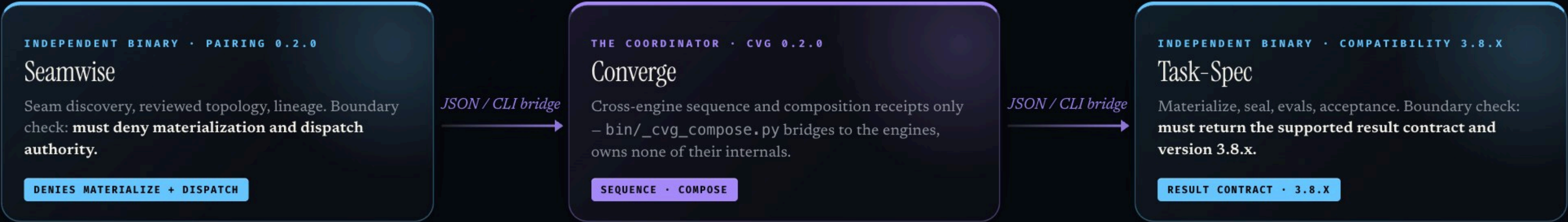
Consent binds the reviewed bytes. **Consent does not transfer to new text.**

HASHED CONSENT · TWO OUTCOMES



Converge composes external engines. It does not vendor them.

Seamwise and Task-Spec remain **independent binaries** with versioned capability contracts — the coordinator speaks to processes over **JSON/CLI bridges**, never to a shared implementation box.



REFUSED **No shared implementation box.** Engines are never vendored into the Converge tree — version pins and capability contracts are the entire interface, so each engine stays independently replaceable.

Compose has five named states and one safe next action.

`cvg compose status` re-hashes the evidence and emits **exactly one** `NEXT=` action — status itself is **read-only**: it writes nothing, it only names the safe move.



STATUS BLOCKS ON

stale review

changed plan

changed task bytes

stale receipt

engine incompatibility

Prepare asks Seamwise for a delivery plan. One named human accepts it.

Two small verbs, one boundary each: `prepare` never touches Task-Spec and never compiles; `review` records one acceptance – **and performs no compilation**.

PREPARE · SEAMWISE ONLY

COMPOSE=NEEDS_REVIEW

cvg compose prepare --source recipe.yaml

BASH

asks

lands at

task-spec

Seamwise for reviewed topology + lineage

NEEDS_REVIEW – awaiting a human

disconnected – not required here

PREPARE DOES NOT

✗ compile a TaskPlan

✗ record an acceptance

✗ create Markdown leaves

REVIEW · ONE NAMED HUMAN

COMPOSE=PREVIEW_READY

cvg compose review --reviewer owner --reason "Topology and rollback accepted."

BASH

signs

moves to

compiler

human card signs the delivery-plan digest

PREVIEW_READY – exactly one acceptance

stays locked – review performs no compilation

REVIEW REFUSES

✗ anonymous reviewer

✗ empty reason

✗ silent acceptance

Preview crosses the engine boundary without writing a single leaf.

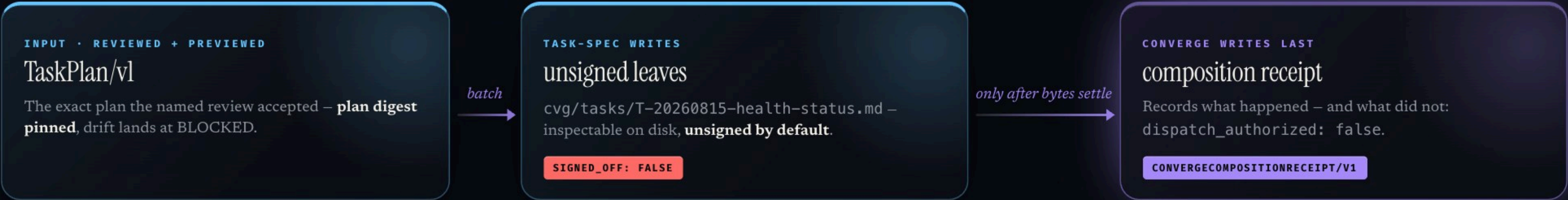
Seamwise compiles `TaskPlan/v1`; standalone Task-Spec validates and previews it – the output is a **read-only proposed DAG**, not files on disk.



PREVIEW FAILS CLOSED
Preview **fails** if the named review is missing or Task-Spec is unavailable. Safe move: record the review – or restore the pinned 3.8.x engine – then rerun `cvg compose preview`.

Materialize writes the unsigned tasks — and the receipt absolutely last.

Task-Spec batches the reviewed plan; Converge writes the composition receipt **only after the exact task set exists** — the pen is held until the bytes settle.



Idempotent INTERRUPTED FINALIZATION recovers cleanly — rerun the same command, no half-written state.	Byte-identical EXACT RERUN same reviewed plan in → same task bytes + receipt out.	14/14 COMPOSE SUITE ROWS review cannot be skipped; tamper and engine drift fail closed.
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The composition receipt is non-authorizing by contract.

It binds engine versions, source commit, lineage, plan digest, task IDs, and task digests – with `dispatch_authorized: false`.

ConvergeCompositionReceipt/v1 – composition receipt

JSON

```
{
  "contract": "ConvergeCompositionReceipt/v1",
  "dispatch_authorized": false,
  "source_commit": "<immutable-source-commit>",
  "task_plan_digest": "sha256:<digest>",
  "task_ids": ["T-20260815-health-status"],
  "task_digests": {"T-20260815-health-status": "sha256:<digest>"}
}
```

FIELD BY FIELD · WHAT THE RECEIPT BINDS	
contract	<i>pins the shape</i> – ConvergeCompositionReceipt/v1
dispatch_authorized	false by contract – <i>materialization never authorizes dispatch</i>
source_commit	<i>the immutable source commit</i> the plan came from
task_plan_digest	sha256 <i>of the exact reviewed TaskPlan</i>
task_ids	<i>the exact task set</i> – one leaf <i>in the demo plan</i>
task_digests	<i>one</i> sha256 per leaf – <i>drift is detectable</i>

NON-AUTHORIZING · READ THIS BEFORE QUOTING A RECEIPT

A valid receipt proves **materialization integrity** – not permission to run, not acceptance after running. Safe next action: authorize one exact revision with `taskspec gate --stamp` (Pass 5, next page).

Materialized means inspectable, not runnable.

Creation never grants authority — open one materialized leaf and **both lifecycle booleans remain false** until Task-Spec itself seals it, and later accepts the work.

TASK-SPEC LEAF • MATERIALIZED

T-20260815-health-status

CVG/TASKS/T-20260815-HEALTH-STATUS.MD

signed_off

false

accepted

false

EXECUTION RAIL

pass 7 selection

execute → locked

HMAC SEAL REQUIRED

UPSTREAM EVIDENCE • SEAMWISE REVIEW

Accepted topology — real evidence, **but it cannot flip the Task-Spec seal.**

UPSTREAM EVIDENCE • CONVERGE MATERIALIZATION

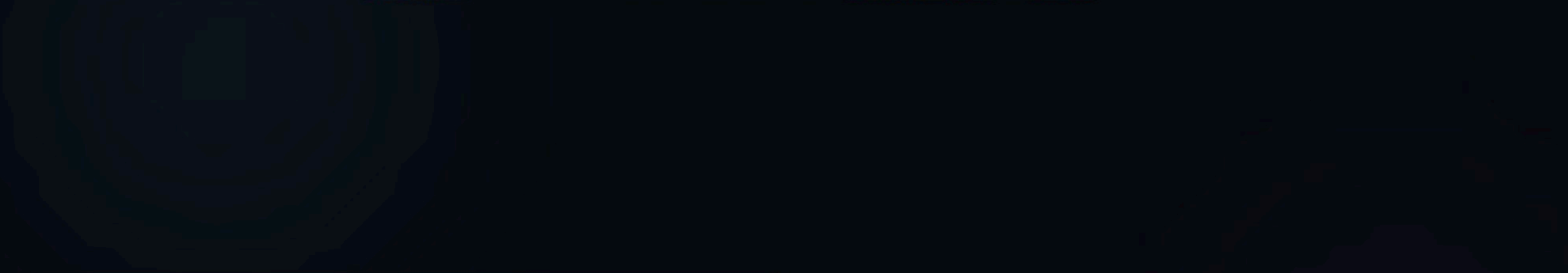
Bytes on disk — real evidence, **but it cannot flip the Task-Spec seal either.**

WHO MAY FLIP WHAT

Only taskspec gate --stamp seals (Pass 5 • P33); only **Task-Spec POST acceptance** sets accepted: true (P53).

Tasking creates the atomic contract — the dry run invents nothing.

Converge offers compatibility doors under `cvg tasks *` and delegates directly to the external engine — `route_taskspec()` in `bin/cvg`.



TASKING · ONE CAPABILITY LEG EXPANDS

CAPABILITY LEG

an accepted, **observable capability state** from the reviewed plan

▼ expands into a small Task-Spec DAG

ATOMIC LEAF

runnable eval

dependency closure included

ATOMIC LEAF

scope + budgets

write scope, iteration ceilings

ATOMIC LEAF

acceptance

criteria the POST gate checks

TASKS PLAN · DRY RUN, NOT A SECOND DECOMPOSER

TASKS_PLAN_TESTS=PASS

```
cv g tasks plan           # dry run · read-only
# derives proposed units from the TaskPlan
# forwards the standalone TaskPlan unchanged
```

BASH

proposed units	<i>TaskPlan rows</i> → exact proposed commands
missing leg	empty yield stays visibly empty — <i>no invented work</i>
mutation	none — <i>read-only door to the external engine</i>

HMAC authorization binds one exact revision.

`taskspec gate --stamp` sets `signed_off: true` only after validating the **exact body digest** under the repository key — one changed byte, and the seal no longer matches.



TRUST LIMIT • WHAT HMAC IS NOT
HMAC is **tamper evidence under a shared key** — it is not **identity** not **secrecy** not **isolation** not **semantic truth**. Anyone holding the repository key can stamp; the seal proves bytes, not wisdom.

Five specs, five issues — and every mismatch glows red.

One signed Task-Spec becomes one tracker issue; `blocked-by` mirrors the DAG; **repository files remain canonical** — dotted lines are a non-authoritative projection.



PARITY VIOLATIONS • PREFLIGHT REFUSES, WRITES NO HALF-BOARD

ORPHAN

tracker issue with **no canonical spec** behind it.

MISSING

sealed spec with **no projected issue**.

CYCLE

blocked-by loops — A blocks B blocks A.

DANGLING

blocked-by points at a **nonexistent issue**.

Pass 6 authors no tasks and changes no dependency truth — it may be skipped for a repo-local queue. Idempotent reruns update in place; receipt stamping preserves the HMAC payload.

Tracker writes are capability-gated and fail-soft.

Identity, projection structure, and tracker mutation are **separate decisions**, backed by private, machine-local state — never by ambient repository files.

CAPABILITY ENVELOPE · REGISTER PROJECTION

tracker.write

an explicit grant is required before any external effect

config

private machine-local state

0600

projection lock

serializes projection writes

0600

symlinked local state

REFUSED

THREE SEPARATE DECISIONS

identity

who the tracker believes this workspace is

projection structure

how specs map onto issues + blocked-by

tracker mutation

the write itself — separately granted

FAIL-SOFT BOUNDARY

Adapter failure is **reported** — it must **not rewrite the task-loop verdict**. Safe move: fix the adapter or its config, rerun the projection; the loop's landing stands.

Bind freezes execution authority — and first proves the input is still the authorized revision.

One signed Task-Spec becomes **one execution profile plus guards, adapters, and a worker brief** — the sealed revision folds into a **7A** runtime contract and a **7B** brief.

THE FOLD · SEALED REVISION → 7A + 7B



INPUT FRESHNESS · FIVE THINGS MUST STILL AGREE

- ✓ **Seal** — the HMAC stamp still validates.
- ✓ **Body digest** — bytes still hash to the pinned revision.
- ✓ **Backend** — the declared runtime is still the one selected.
- ✓ **Evidence** — the cited slice is present and current.
- ✓ **Task identity** — the same task, even moved to `tasks/done`.

RED GATES · REFUSAL + SAFE MOVE

- ✗ **Mutated body** — binding refused; re-seal the exact revision, then re-bind.
- ✗ **Runtime drift** — selection refused; re-run bind against current capabilities.

READ-ONLY RE-VERIFICATION
`bind --check` re-verifies without writing, and a task moved to `tasks/done` remains verifiable by identity — **a stale or unsigned task never reaches runtime selection.**

The profile carries enforcement, not a second copy of the task.

Pin the task revision, backend, topology, budgets, evidence slice, path policy, capabilities, and authority epoch – **the profile references the task; it never becomes a second task contract.**

ANNOTATED SPECIMEN • ONLY LOAD-BEARING FIELDS

cvlg/execution/T-20260815-health-status.profile.yamlYAML

```
# execution profile – emitted by cvlg bind • 7A contract
task:      T-20260815-health-status
revision:   sha256:<digest>    # sealed body – pinned exact
backend:    codex              # one declared runtime
topology:   single            # default – parallel earns it
budgets:
  iterations: 15              # signed ceilings – tighten only
  wall_seconds: 5400
  tokens:     120000
evidence:   cvlg/evidence/...  # slice the loop may cite
path_policy: repository-gate  # inner fence – outranks scope
capabilities:
  grants:     []              # default deny
authority_epoch: <stamped-at-seal> # re-seal bumps the epoch
```

WHY THESE EIGHT FIELDS

revision	re-hashed at selection – mutation turns the gate red
topology	single by default; parallel must earn it (P40)
capabilities.grants	[] = default deny – nothing implicit
authority_epoch	re-sealing bumps the epoch; stale profiles fail closed

PROFILE CHECKS • RUNTIME SUITE

THINPORTABLEDETERMINISTIC

Thinner than the Task-Spec on purpose – it carries what the runtime enforces, nothing more.

BOUNDARY • ONE CONTRACT ONLY

Prose stays in the sealed Markdown. The profile carries only what the runtime enforces – **a reference to authority, not a rival source of it.**

The repository gate outranks a task that asks for too much.

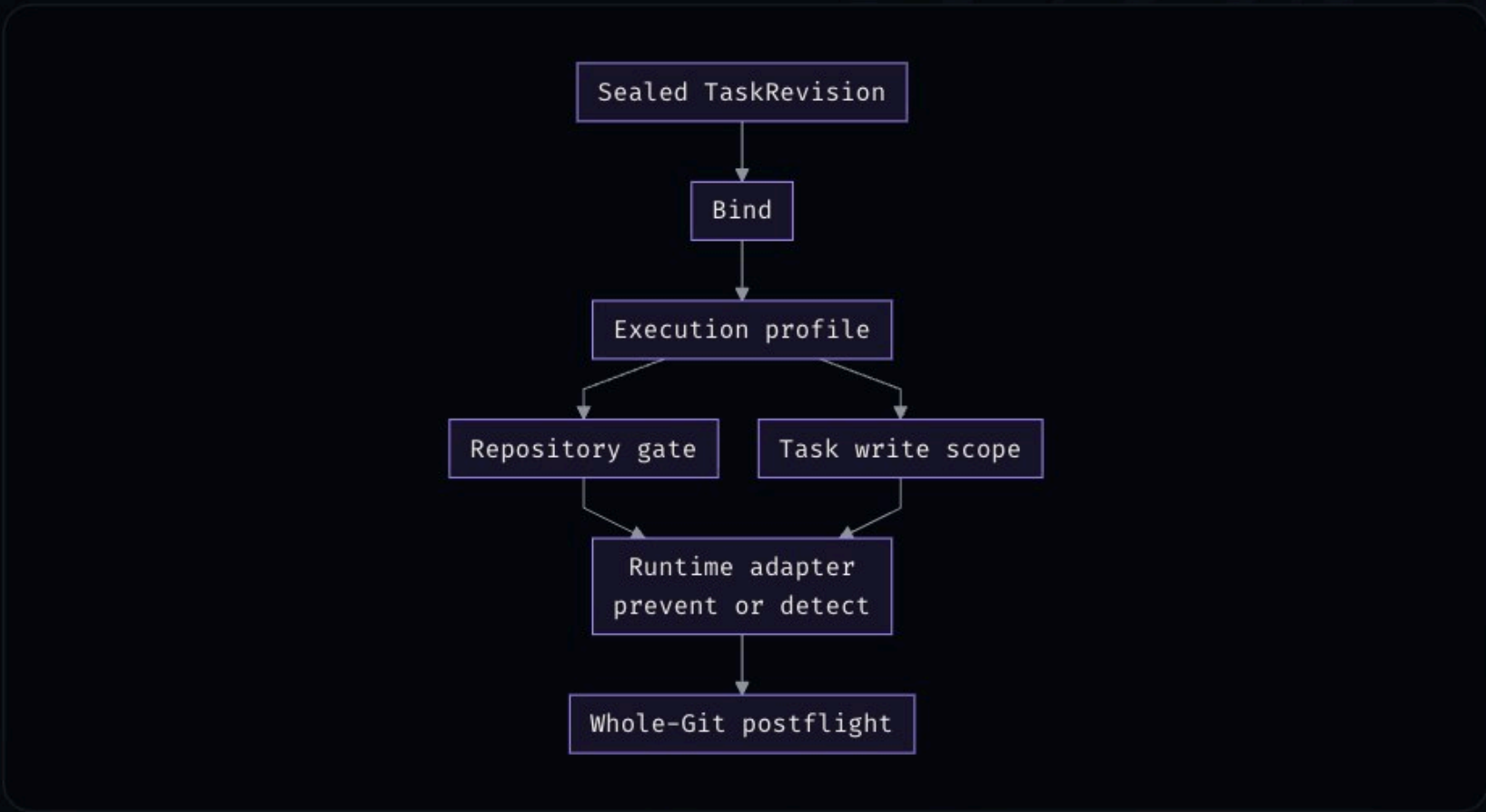
The task authorizes a **candidate set**; the trusted repository policy may **narrow** it — and the fence protects itself from deletion or weakening.

TWO FENCES • CANDIDATE SET VS TRUSTED POLICY



task scope is necessary — the repository fence can still be stricter

CANONICAL DIAGRAM D • BINDING FENCES (VERBATIM)



NESTED PROJECTS

Inherit the Git-root policy

A project nested inside the repository does not escape the root fence — the trusted policy still applies.

RENAMES

Both endpoints checked

A rename is evaluated at **both endpoints** — source and destination — never just one side.

WHOLE-REPO DIFF

Sibling changes caught

The gate diffs the **whole Git reality**, so a change beside the declared write set cannot slip through.

INVALID POLICY

Fails closed

An unreadable or invalid policy **refuses the gate** — fix the policy file, then re-run the check.

Prevent and detect are different control classes.

Codex may **prevent** some writes before the tool call; Claude and Kimi often **detect** violations postflight — the resolver manifest records which class you actually got.



RESOLVER MANIFEST · RECORDED, NOT MARKETING

ADAPTER	RECORDED CLASS	WHERE IT BITES
codex	prevent · some writes	before the tool call
claude	detect · postflight	at the gate, after the fact
kimi	detect · postflight	at the gate, after the fact

NEVER-CLAIM TRAP

Documentation may not upgrade **detect** to **prevent**.

The manifest records the class each adapter actually enforces. Safe move: record the measured class — and when prevention is required, place the write behind the repository gate instead of trusting the prose.

Parallel must earn disjoint write sets. The brief never rewrites the plan.

Single is the default; intra-task parallel requires substantive evidence and **disjoint ownership** — and 7B emits `AGENTS.task.md`, a pointer back to the sealed revision.

INTRA-TASK TOPOLOGY · ONE TASK, ONE OR MORE WORKERS

SINGLE · THE DEFAULT

One worker, one bounded attempt

The sealed task runs as one worker against one write set — **no extra evidence required**.

PARALLEL · MUST BE EARNED

Disjoint ownership, or stay single

Requires **substantive evidence**, and the write sets must not overlap.

worker A · write set A

$n = \emptyset$

worker B · write set B

Topology never creates new task authority or cross-task scheduling.

7B · WORKER BRIEF — LINKED BACK TO THE SEALED REVISION

AGENTS.task.md — emitted by 7B MD

```
# worker brief — names, never re-plans
task:      T-20260815-health-status
epoch:     <authority-epoch>
scope:     <sealed write scope>
proof route: tier-1 eval → path gate
source:    cvg/tasks/T-20260815-health-status.md @ sha256:<digest>
```

*Identifiers, epoch, scope, proof route, source contract — **not a rewritten implementation plan**.*

STALE BRIEF · REFUSED

A stale brief **fails the gate** — re-bind to refresh it against the current revision.

ROUTER SETUP · NON-CLOBBER

Generic router setup **proposes beside human content** — it never clobbers it.

Adapters are execution bridges, not new authorities.

One profile feeds three adapter sockets; each selects a declared runtime, **closes stdin**, **filters the environment**, and reports what it can enforce.



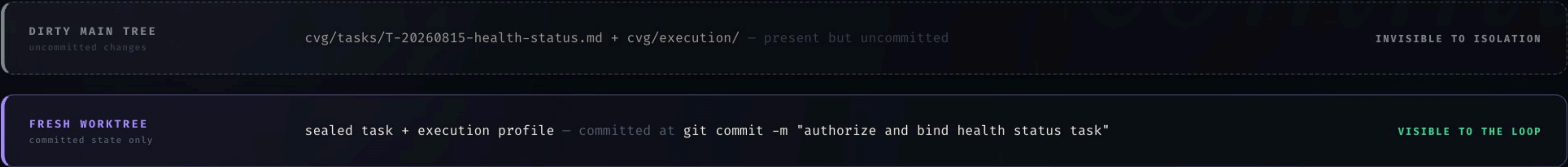
ALL THREE RETURN TO THE SAME VERIFICATION RAIL TIER-1 EVAL + PATH GATE

REFUSAL · SAFE MOVE

An explicit **engine/profile mismatch fails before execution** — align the declared runtime with the profile, then re-bind.

Commit the contract, then re-verify it without writing a byte.

Worktree isolation sees **committed state only** – the sealed task and execution profile must be committed before the loop, or the fresh checkout cannot see them.



Committing the contract is *not* merging product work or opening a PR.

LAST PREFLIGHT · FIVE INPUTS CONVERGE, READ-ONLY



One named issue enters one loop cell. The loop never picks its own work.

`cvb loop --issue ...` receives **one assignment**. Manager/fleet scheduling remains **outside v0.2.0**.

V0.2.0 PRODUCT BOUNDARY

`cvb loop --issue T-20260815-health-status --agent codex # one assignment`

BASH

1 ONE LOOP CELL

bounded attempt cycle

Checkpoint → attempt → verify → landing. **Finite budgets**, one issue in, one named landing out – no queue scan, no fan-out.

ONE ISSUE IN • ONE LANDING OUT

OUTSIDE THE BOUNDARY • RANK F – ROADMAP

- MANAGER • FLEET SCHEDULING
- AUTONOMOUS FAN-OUT
- FRONTIER SELECTION

*A future fleet silhouette – **never rendered as a current capability** in v0.2.0.*

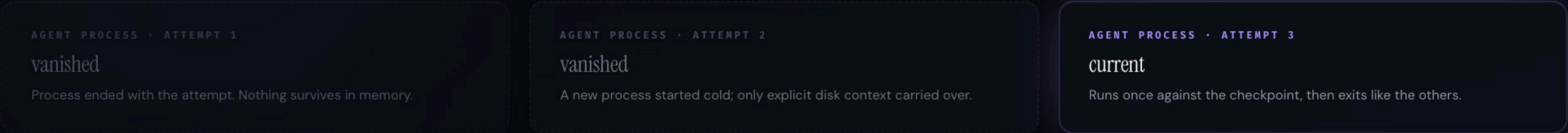
BOUNDARY • REFUSAL

No issue ID is a **usage error**, not permission to scan a queue: Pass 8 may not invent or fan out tasks. Safe move – name exactly one issue.

Each retry is a fresh agent process. Continuity lives on disk.

Each retry gets a new agent process — the **spec, diff, Git history, checkpoint, and attempt log** carry continuity.

EPHEMERAL · ONE AGENT PROCESS PER ATTEMPT, EXITS WITH THE ATTEMPT



previous-failure context is **bounded and explicit** — handed over on disk, never in chat memory

DURABLE · DISK ARTIFACTS PERSIST BELOW THE PROCESS LINE



BOUNDARY
Hidden chat memory is **not loop state** — the loop trusts only what it can re-read from disk.

Iterations, working time, and tokens are independent ceilings.

A CLI flag may **tighten** the signed maximum – it may **never raise** it.

HARD SIGNED CEILING • SEALED IN THE TASK-SPEC

a CLI flag tightens below this line – never above it

ITERATIONS

15 attempts



Counted attempts. Reaching the ceiling lands **EXHAUSTED** – exit 1, handoff and worktree preserved.

WORKING TIME

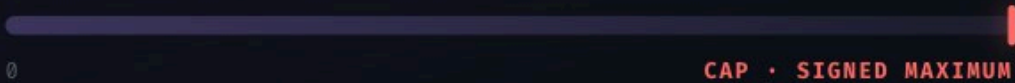
5400 wall_seconds



--resume continues **accumulated working time** – not wall time spent paused.

TOKENS

120000 tokens



A missing token budget is reported as **unbounded** – never guessed.

THREE BUDGETS FAIL DIFFERENTLY

Each ceiling lands its own named stop – a spent budget is **EXHAUSTED** (exit 1), never a silent success and never a reason to widen the seal.

The lane turns the cost dial. It never touches the signed ceilings.

FAST does not draw the largest model; FULL enables tier 2; **effort scales per-attempt time**.

LANE × EFFORT MATRIX → MODEL TIER · REASONING · TIMEOUT · VERIFICATION

LANE	MODEL TIER	REASONING	PER-ATTEMPT TIMEOUT	VERIFICATION DEFAULT
FAST	lighter tier — never the largest	reduced	shortest per attempt	tier 1 — may enter at Pass 5
NORMAL	standard tier	standard	standard per attempt	tier 1 — skips optional ceremony
FULL	largest tier	extended	longest per attempt	tier 2 enabled by default — explicit <code>--no-verify</code> may override

NORMAL/FAST default to local settlement without tier 2 · sensitive/greenfield floors can force a stronger lane · effort scales per-attempt time inside every lane

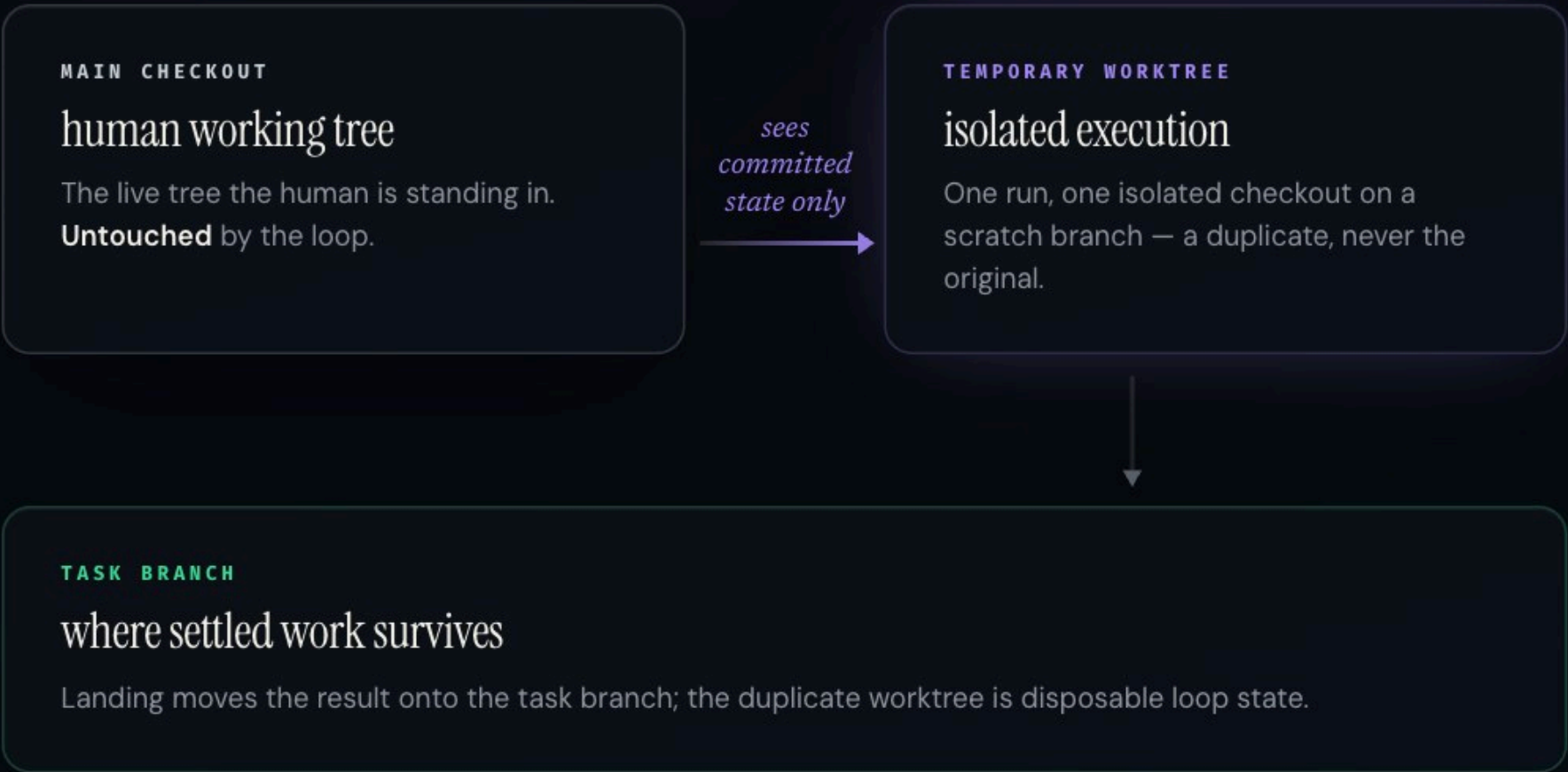
BOUNDARY · REFUSAL

No lane widens a signed budget or waives a gate. `--no-verify` is an explicit override, never a default — safe move: raise assurance by choosing a stronger lane, not by loosening the seal.

One run gets one isolated checkout. Your tree stays untouched.

One isolated checkout and one scratch task branch per run — **the human’s working tree remains untouched.**

DEFAULT SAFETY POSTURE • ISOLATION: WORKTREE



LANDING CLEANUP RULES

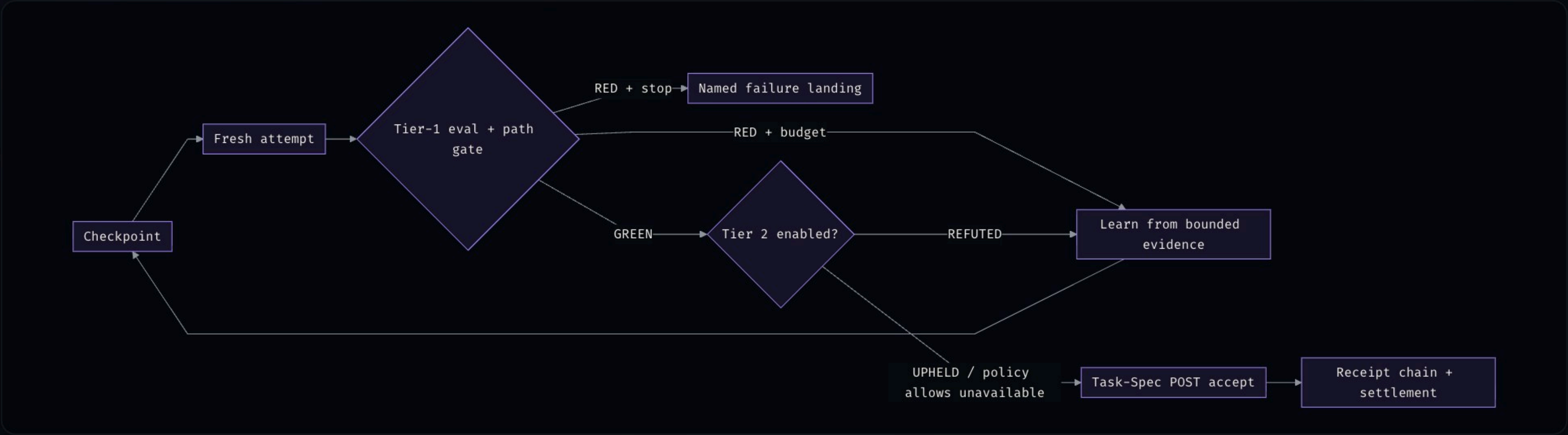
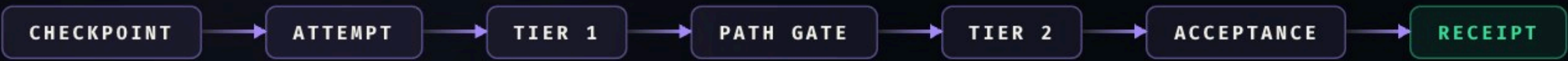
LANDING	DUPLICATE WORKTREE
SETTLED LOCAL_SETTLED	cleaned — settled work survives on the task branch
BLOCKED STALLED EXHAUSTED	kept — uncommitted work stays inspectable
failed empty run	cleaned up — nothing worth keeping

Isolation is the default posture, not an opt-in flag: **isolation: worktree** in the §8 loop policy.

Attempt, verify, then learn or settle.

Run the worker once, run the sealed eval, inspect path policy, optionally refute – then **settle or create the next bounded attempt**.

ON-SLIDE ORDER



STAGNATION BREAKER • THE SECOND BRAKE

Repeated ineffective attempts land **STALLED** (exit 1) before the full budget is burned – the ineffective engine stops at **three attempts** rather than the declared fifteen.

loop kernel 57/57

Eight landings. Only three exit zero.

Success, no work, human dependency, stagnation, exhaustion, cancellation, and coordinator error **never collapse into one status**.

SETTLED · LOCAL_SETTLED · NO_OP → exit 0 · BLOCKED · STALLED · EXHAUSTED · CANCELLED → exit 1 · ERROR → nonzero, a safe failure

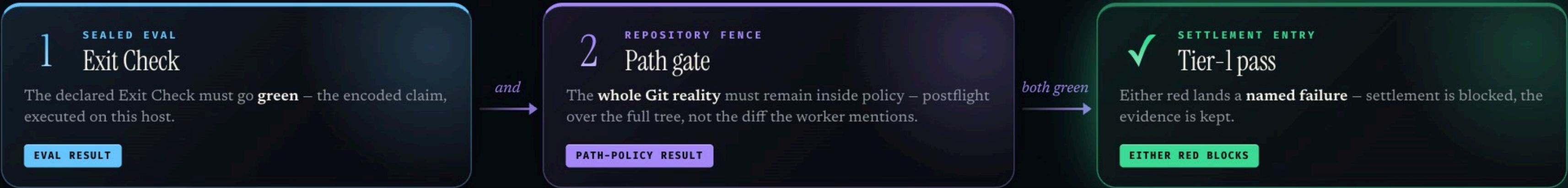
SETTLED accepted; external publication allowed EXIT 0 TASK BRANCH + RECEIPTS	LOCAL_SETTLED accepted; external writes denied EXIT 0 LOCAL TASK BRANCH + RECEIPTS	NO_OP already green at entry EXIT 0 EXISTING STATE	BLOCKED human/upstream/tier-2 decision needed EXIT 1 HANDOFF; INSPECTABLE WORKTREE WHEN APPLICABLE
STALLED stagnation breaker fired EXIT 1 HANDOFF + WORKTREE	EXHAUSTED signed budget reached EXIT 1 HANDOFF + WORKTREE	CANCELLED external stop signal EXIT 1 WRITTEN CHECKPOINT/HANDOFF	ERROR coordinator cannot continue safely EXIT ≠0 ONLY TRUSTWORTHY EVIDENCE RETAINED

OPERATIONAL FAILURE · NOT COLLAPSE

Every stop has a name and an exit code – a spent budget, a fired breaker, or a cancellation is **reported, never converted into success**.

The eval must go green — and the whole Git reality must stay inside policy.

Tier 1 runs the sealed eval and the repository fence: the two results **converge**, and either red blocks settlement.

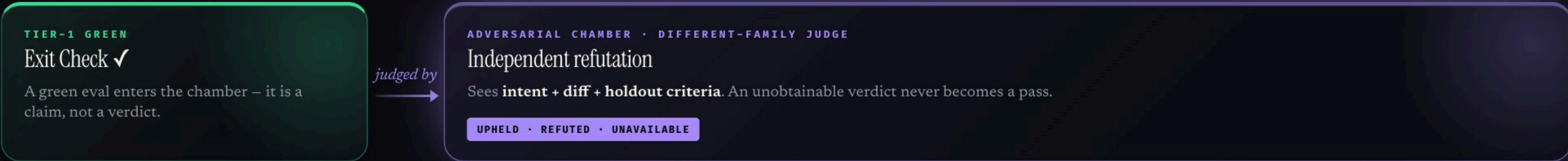


BOUNDARY • TRUST LIMIT

A zero exit proves only **the encoded claim on this host** — it can be weak or gamed. Safe move: keep the path fence on, and let higher-risk lanes add a tier-2 refutation.

Tier 2 tries to refute a green result.

The judge sees **intent, diff, and holdout criteria** — and returns UPHELD, REFUTED, or unavailable.



UPHELD

the claim survives refutation — proceeds to Task-Spec POST acceptance

REFUTED

loops back — learn from bounded evidence, then a fresh checkpoint

UNAVAILABLE

never a pass: low-risk documented unavailability may proceed under policy; high-risk work stops

TRUST LIMIT • HOLDOUT HONESTY

The judge is intended to own **## Holdout** — **but the worker brief still points at the Task-Spec that contains it.** Refusal: never claim the worker could not read the evaluator criteria. Safe statement: holdout secrecy is a **goal, not a shipped invariant** — treat tier 2 as defense-in-depth, not a hidden exam.

Settlement is local first. External effects stay locked.

Without explicit `external_writes: allow` and capability grants, the loop **settles locally** and suppresses tracker/remote effects.

WHAT HAPPENS BY DEFAULT

LOCAL TASK BRANCH accepted work landed	branch + commits in place	WRITTEN
SETTLEMENT RECEIPT hash-linked claim	LOCAL_SETTLED recorded	WRITTEN
PULL REQUEST external effect	not opened	LOCKED
TRACKER / REMOTE external effect	mutation suppressed	LOCKED

default-deny loop policy - §8

YAML

```
policy:
  external_writes: deny
capabilities:
  grants: []
```

SKIPPED / FAILED SEMANTICS

Unauthorized tracker writes report **SKIPPED**; adapter failure reports **FAILED** – neither changes the loop landing.

Only **ACCEPTED=1** closes acceptance.

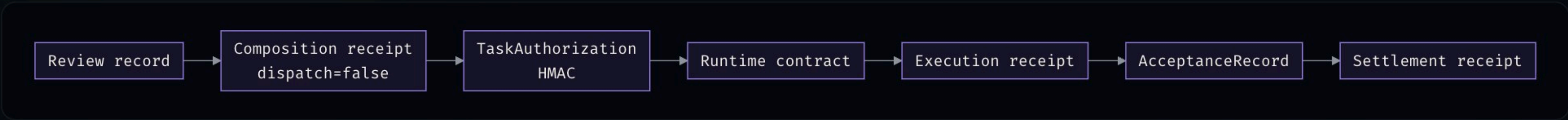
Converge hands the **exact revision and attempt** back to Task-Spec; the AcceptanceRecord returns.



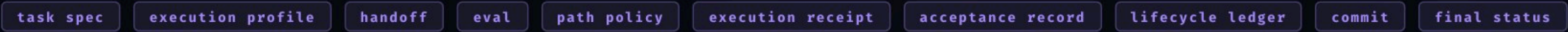
BOUNDARY · REFUSAL
Converge does not set `accepted: true` itself – **worker narration and the execution receipt are insufficient**. Safe move: route the exact handoff through the Task-Spec POST gate.

Settlement is a hash-linked claim, not a victory message.

Ten records must agree; the success receipt is written **after settlement**, never before.



TEN RECORDS MUST AGREE



PROOF • CLEAN ROOM

The clean-room test verifies the **hash chain and lifecycle ledger agreement** – 21/21 rows, end to end: copy install → init → seal → bind → loop → accept → done.

cvg next reads the floor. It does not remember state.

“Artifacts reveal the current boundary. The conductor names the owning skill and gate; **the gate still decides.**”

THE FLOOR • CANONICAL CVG/ TREE

cvg/
├ tasks/ unsigned + sealed leaves • digests
└ execution/ attempts • receipts

Every run re-reads the artifacts fresh – the conductor holds **no memory** between runs.

EVIDENCE PRESENCE IS NOT VERDICT

A stale Pass 4 hash displays **[!]** and blocks later passes – the conductor **refuses Pass 5 behind stale consent**. Safe action: **prepare / review again** to refresh the digest.

EVIDENCE BOARD • READ-ONLY PROJECTION

PASS	EVIDENCE ON THE FLOOR	OWNING SKILL	CLOSING GATE
0 • Capture opt	BRD or no-go	idea-to-brd	CHECK_BRD=PASS
1 • Intent	signed BRD → falsifiable tech-spec	brd-docs-to-tech-req	CHECK_TECH_SPEC=PASS
2 • Structure	ADRs + glossary	tech-req-to-adrs	CHECK_ADR=OK
3 • Decompose	swimlane/leg tree	reqs-to-swimlane-plans	CHECK_PLAN=OK
4 • Consensus 	hardened plans • review record — digest stale [!]	sketch-plans-adversarial-review	CHECK_CONSENSUS=OK stale
5 • Tasking	HMAC-sealed atomic DAG	taskspec 3.8.x external	TIER=1
6 • Register opt-in	1:1 tracker projection	task-specs-to-issues	CHECK_REGISTER=OK
7 • Bind	profile + guards + brief	task-to-runtime-contract	CHECK_RUNTIME_CONTRACT=PASS
8 • Loop	named landing + evidence	task-loop	TASK_LOOP=* → ACCEPTED=1

[!] stale Pass 4 digest – Pass 5 refused behind stale consent; later passes blocked until re-review

NEXT_PASS=N | DONE one token, recomputed from the floor on every run – the conductor remembers nothing

Guided chat adds four choices — then waits.

“Uncommitted `cvg next --guided` adds `CONTINUE`, `EXPLAIN`, `INSPECT`, and `PAUSE`, then waits.”

W

NOT COMMITTED

GUIDED CHOICE 01 · ROUTE

CONTINUE

Move to the named next pass. The owning skill runs and the pass gate still decides — the chat only routes the human.

GUIDED CHOICE 02 · READ

EXPLAIN

Hear why the board looks like this: which artifacts exist, which gate is open, and what evidence is still missing.

GUIDED CHOICE 03 · OPEN

INSPECT

Open the exact artifact paths behind a row — you read the evidence itself, never a chat reconstruction of it.

GUIDED CHOICE 04 · STOP

PAUSE

Stop the session in place. Nothing is written and nothing is remembered — silence never means continue.

THE SAME EVIDENCE BOARD — UNCHANGED

CVG/ ARTIFACTS

CONDUCTOR READS

PASS GATE DECIDES

NEXT_PASS=N | DONE

No chat state is persisted; silence never means continue. Pre/post hooks and pass gates remain authoritative.

Every public form speaks one machine envelope.

“Global `--json` and `--dry-run` are position-independent; **exit code, token, mutation truth, and error remain explicit.**”

SPECIMEN · CONVERGECLIRESULT/V1

cv g compose preview --json · exit 0JSON

```
{
  "contract": "ConvergeCLIResult/v1",
  "ok": true,
  "token": "COMPOSE=PREVIEW_READY",
  "verdict": "PREVIEW_READY",
  "exit_code": 0,
  "changed": false,
  "dry_run": false,
  "data": {},
  "error": null,
  "meta": {"schema_version": 1, "cvg_version": "0.2.0"}
}
```

THE ENVELOPE, FIELD BY FIELD

FIELD	RULE
contract	names the schema — ConvergeCLIResult/v1
ok	did the contract hold — boolean, never implied
token	stable machine token — COMPOSE=PREVIEW_READY
verdict	human-readable twin of the token
exit_code	process truth — 0 success · 1 refusal · nonzero safe failure
changed	mutation truth — did bytes move (preview: false)
dry_run	rehearsal flag — a dry run mutates nothing
data / error	payload object · explicit error or null
meta	schema_version 1 · cvg_version 0.2.0

60

FORMS · CHECKOUT MATRIX

W CHECKOUT ONLY

239

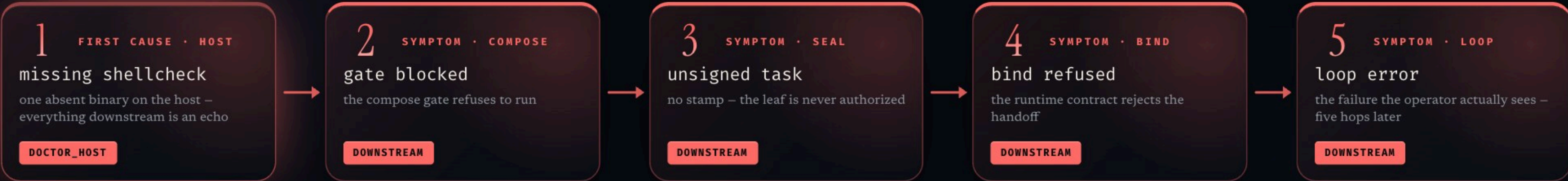
CALLS COVERED

25/25

JSON ENVELOPE ROWS

Five symptoms, one broken hop. The doctor names the first.

“Host, evidence, plugin, runtime-contract, and setup checks turn distant symptoms into **one actionable upstream cause**.”



FIVE CHECK FAMILIES · ONE REPORT

DOCTOR_HOST

DOCTOR_EVIDENCE

DOCTOR_PLUGIN

CHECK_RUNTIME_CONTRACT=PASS

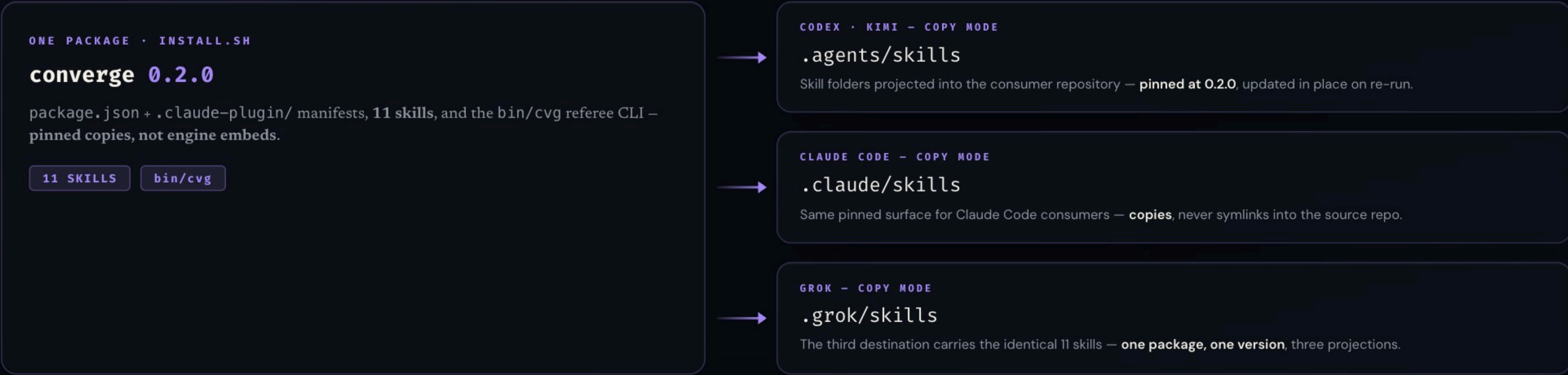
SETUP CHECKS

REFUSAL · DO NOT PATCH FIVE SYMPTOMS

The chain has **one upstream cause**. Safe action: run the Converge doctors, fix the named first cause – **install shellcheck** – then re-run the gate. The chain unblocks at its first hop.

One package, three homes. The engines stay outside.

“Copy mode pins **11 skills and the CLI** into Codex/Kimi, Claude Code, and Grok destinations; Task-Spec and Seamwise remain external.”



ENGINES REMAIN EXTERNAL — NEVER EMBEDDED

taskspec · TASK-SPEC 3.8.x

seamwise · 0.2.0

binaries live outside the consumer; skills invoke them by contract.

LOCAL CONFIG POSTURE

0600 CONFIG

SYMLINK-SAFE

IDEMPOTENT

GIT-EXCLUDED

Eleven skills route intent. Pass 5 stays outside.

“Eight pass skills, three utilities, and **no mirrored Task-Spec skill**.”

THE PACKAGE • 11 TILES • DASHED = OPTIONAL HOP

PASS 0 • CAPTURE • OPTIONAL
idea-to-brd

PASS 1 • INTENT
brd-docs-to-tech-req

PASS 2 • STRUCTURE
tech-req-to-adrs

PASS 3 • DECOMPOSE
reqs-to-swimlane-plans

PASS 4 • CONSENSUS • BARRIER ⓘ
sketch-plans-adversarial-review

PASS 6 • REGISTER • OPT-IN
task-specs-to-issues

PASS 7 • BIND
task-to-runtime-contract

PASS 8 • LOOP
task-loop

UTILITY • CONDUCTOR
evidence-to-next-pass

UTILITY • LESSONS
pass-to-lesson

UTILITY • AUTHORING
skill-creator

PASS 5 • TASKING
– no skill here –

EXTERNAL ENGINE • NOT A SKILL

taskspec 3.8.x

Pass 5 Tasking – HMAC seal, handoff, eval, acceptance – is **invoked by contract**, never vendored into the package.

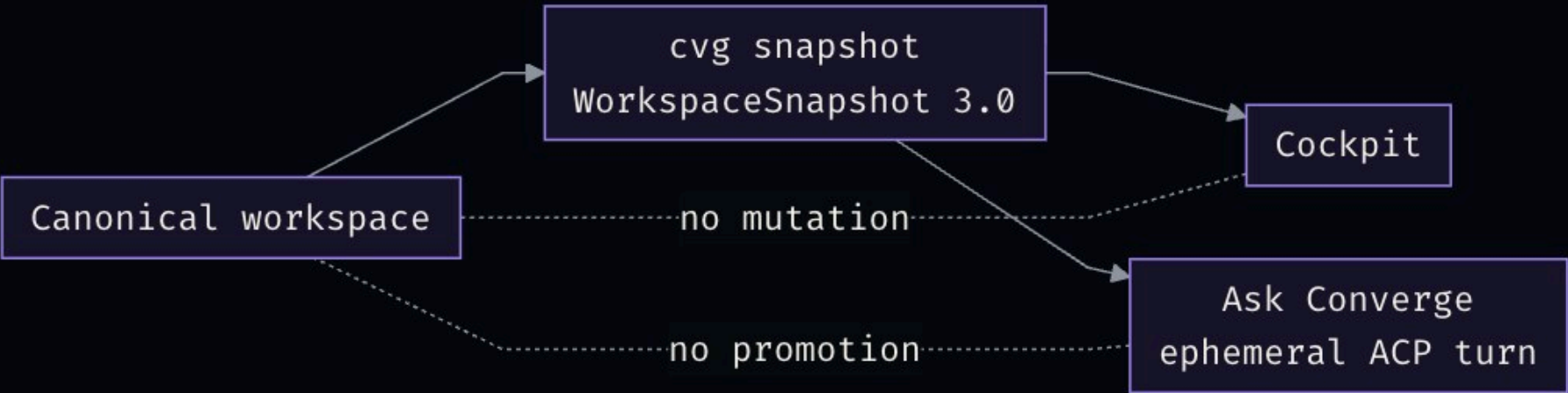
PACKAGE TRUTH • OBSERVED

version unity – **exactly 11 skills at 0.2.0**
package – **244 files** • no vendored engine tree

PACKAGE=READY files=244 skills=11

Cockpit reads one snapshot. No arrow leads back.

“The UI renders descent, decomposition, gates, queue, attempts, receipts, health, and documents from `WorkspaceSnapshot 3.0`.”



OBSERVES • EIGHT SURFACES

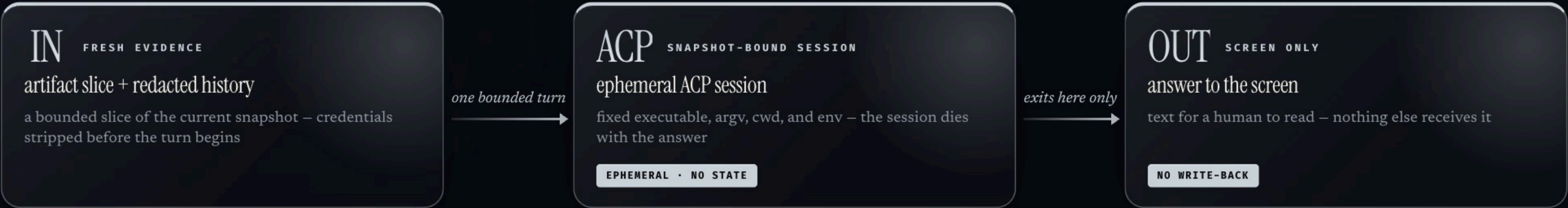
- descent
- decomposition
- gates
- queue
- attempts
- receipts
- health
- documents

SECURITY POSTURE • ENFORCED BY THE COCKPIT GATE

- loopback only
- explicit roots
- token + exact-origin transport
- GET-only observation
- snapshot+SHA artifact allowlist
- traversal + symlink rejection
- binary + credential rejection

Ask explains the evidence. It cannot touch it.

“One snapshot-bound ACP turn can **explain** artifacts; it cannot create, approve, bind, transition, or settle them.”



NEVER FROM ASK · INTERPRETATION IS NOT PROOF

× CREATE × APPROVE × BIND × TRANSITION × SETTLE

Gates, seals, and receipts remain the only writers – Ask only reads.

BOUNDED BY CONSTRUCTION · CONTROLS

fixed executable / argv / cwd / env permission denial unsafe-tool refusal

bounded history + output cancellation stale-turn rejection credential redaction

Recovery follows tokens, not optimism.

“Compose status emits one `NEXT=`. Loop landings preserve **the handoff and work that a human needs.**”

ONE ARROW PER STATE · THE RECOVERY VOCABULARY

STATE	WHAT THE SIGNAL MEANS	ONE SAFE NEXT ACTION
stale review COMPOSE	the recorded review digest no longer matches the current topology	<code>prepare</code> / <code>review</code> again to refresh consent
interrupted materialize COMPOSE	finalization stopped mid-write; no half-completed receipt stands	rerun idempotently — an exact rerun is byte-identical
EXHAUSTED <code>EXIT 1</code> LOOP · SIGNED BUDGET REACHED	budget spent before landing; handoff + worktree preserved	raise the signed budget — or split the task
REFUTED TIER 2 · ACCEPTANCE	independent acceptance rejected the work	fix the work, then resubmit to the same gate
CANCELLED <code>EXIT 1</code> LOOP · EXTERNAL STOP SIGNAL	the operator stopped the loop; written checkpoint/handoff preserved	resume from the handoff

GAP · CURRENT --RESUME

Under worktree isolation, **--resume starts a fresh tree at attempt 1**. Never imply continuation of uncommitted work — recovery resumes from the handoff, not from the old tree.

Fifty-seven committed forms; sixty only in the working tree.

The committed 0.2.0 main surface has **57 forms**. The audited dirty checkout adds three guided/conductor forms – **and tests 60**.

COMMITTED CLI MATRIX · CONTRACTS/CLI-COMMAND-MATRIX.JSON @ 58B1DDB · ONE TILE PER FORM

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38
39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57

W CHECKOUT ONLY

audited dirty checkout +3 guided/conductor forms – matrix expands 57 → 60 · JSON matrix 60 forms / 239 calls · conductor checkout 44/44 – observed in the dirty tree only

GUIDED CONTROL

cvg next --guided

Adds **CONTINUE · EXPLAIN · INSPECT · PAUSE** over the same evidence board — then waits. Silence never means continue.

CONDUCTOR HOOK

cvg next pre N

Pre-pass routing hook. Pre/post hooks and pass gates remain authoritative — the checkout adds **no new authority**.

CONDUCTOR HOOK

cvg next post N

Post-pass routing hook. No chat state is persisted; canonical state stays in the **cvg/** artifacts.

BOUNDARY · CLAIM DISCIPLINE

Never render 60 as released – not until the guided-chat change set is committed and release evidence is refreshed. The safe claim: **57 committed forms** on main; 60 tested in the audited checkout.

The repository is an executable method, not one monolith.

CLI, contracts, skills, observer, tests, scripts, evidence, templates, and assets have **distinct jobs** – and the generated views never outrank their sources.

508

TRACKED PATHS • AUDITED 2026-08-26 @ 58B1DDB

templates ship inside `skills/` • `assets/` holds 21 brand + process files

bin/ referee CLI + helpers – the command surface 7	contracts/ CLI matrix + versioned JSON Schemas 6	skills/ 11 packaged skills – scripts, tests, templates 220	apps/ Cockpit observer – client/server, Ask, schemas 102
tests/ aggregate + subsystem gates 22	scripts/ bootstrap, docs, evidence, release, demo 11	evidence/ retained receipts, snapshots, releases 72	docs/ authority, trust, guides, generated CLI reference 18

SOURCE OF TRUTH → GENERATED VIEW – NEVER THE REVERSE



v0.2.0 is immutable. Main is 42 commits later. Guided chat is later still — and uncommitted.

A release freezes one corridor inside a moving repository — **every claim must name the corridor it stands on**.

RELEASE • V0.2.0 frozen tag • immutable evidence	3de9f0b • 3de9f0b5f83f1bb62475308317c58e53f851b0db release workflow + archived assets live on this tag	IMMUTABLE
CURRENT MAIN origin/main • committed surface	58b1ddb • 42 commits after the tag 57 committed CLI forms • audit baseline 2026-08-26	COMMITTED
WORKING TREE audited checkout • uncommitted	58b1ddb + diff • 16 modified + 1 untracked 60 forms tested • guided-chat change set not committed	W CHECKOUT ONLY

REPORTED HOSTED EVIDENCE • DOCUMENTARY RANK [D]

README records all eight release jobs green on feature SHA 1fa0545... in run 32048296517. Treat as repository-reported unless refreshed live — hosted state drifts; the tag does not.

Hundreds of discriminating rows. No inflated total.

Direct 2026-08-26 runs prove named behaviors across passes, composition, runtime, settlement, installation, JSON, and Cockpit – **grouped by subsystem, never summed.**

REGISTER • PASS 6

145

145/145 — parity, idempotency, DAG, receipt stamping, identity, projection.

TESTS/TEST-REGISTER.SH

LOOP KERNEL • PASS 8

57

57/57 — brakes, budgets, isolation, tier 2, path + external-write policy.

TESTS/TEST-LOOP-KERNEL.SH

RUNTIME CONTRACT • PASS 7

48

48/48 — policy, binding, freshness, topology, adapters, settlement.

RUNTIME CONTRACT SUITE

PASSES 0-4 • STRUCTURAL GATES

39+24+38+33+23

Capture, Intent, Structure, Decompose, Consensus rows discriminate their named fixtures.

SKILL PASS-GATE SUITES 0-4

CLEAN ROOM • E2E

21

21/21 — copy install → init → seal → bind → loop → accept → done.

TESTS/TEST-CLEAN-ROOM-INSTALL-E2E.SH

COMPOSE

14

14/14 — review cannot be skipped; tamper and engine drift fail closed.

TESTS/TEST-COMPOSE.SH

JSON CONTRACT

W CHECKOUT

60/239

60 forms / 239 calls serializable in the checkout matrix; committed envelope 25/25.

TEST-CVG-JSON-MATRIX.PY • TEST-CVG-JSON-ENVELOPE.SH

COCKPIT GATE

102

40 server + 62 client — read-only bridge, bounded Ask, snapshots; production build green.

APPS/COCKPIT SUITES

BOUNDARY • NO FAKE UNIVERSAL COUNT

These suites overlap by design — a row in one may exercise another's surface. **No grand total appears on this page**, because summing them would manufacture a number no run ever produced.

The tracked root gained two files. The layout allowlist did not.

tests/test-repo-layout.sh reports tracked AGENTS.md and OPERATING.md as undeclared top-level files – the aggregate exits 2 at the first gate.

AGGREGATE MAKE CHECK ON CURRENT MAIN • PHASE ORDER



EXACT FAILURE • SAFE REMEDIATION

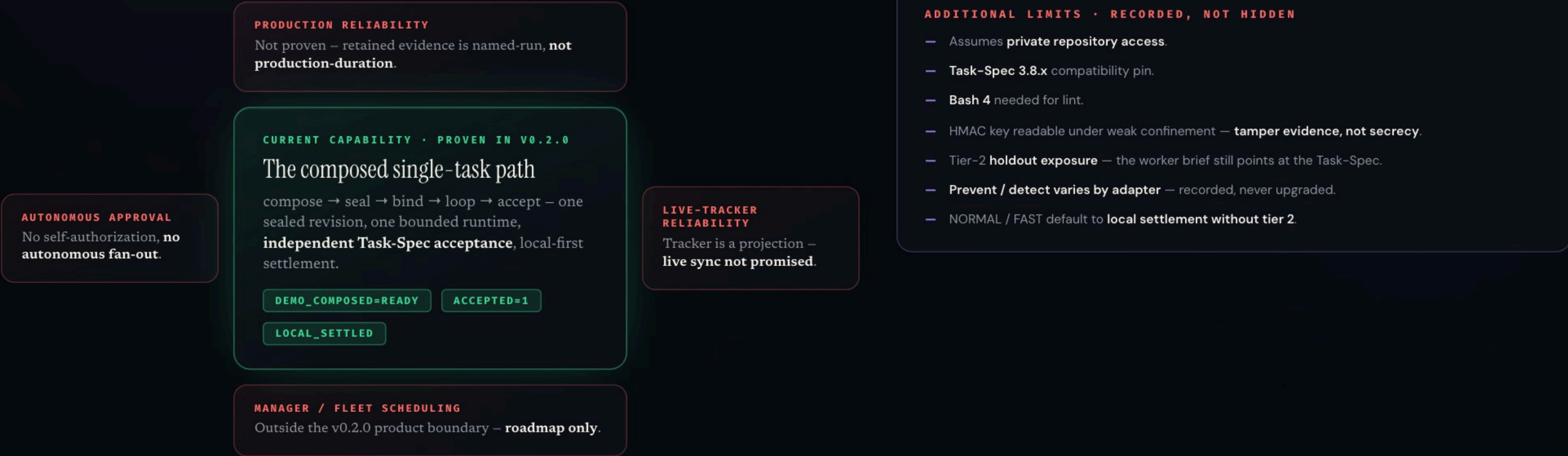
Aggregate **make check** exits 2 before the green phases run.

tests/test-repo-layout.sh rejects tracked AGENTS.md + OPERATING.md as undeclared top-level files; the JSON, docs, compose, and package phases **pass when run independently**, but the aggregate never reaches them. **Safe remediation:** update the declared layout contract, or relocate the files through an explicit repository decision, then rerun the full aggregate gate – **release-ready may not be claimed**.

v0.2.0 proves a composed single-task path, not a factory fleet.

Reproducible composed execution is real. Production reliability, autonomous approval, live-tracker reliability, and Manager scheduling are **not promised**.

CAPABILITY ISLAND · SURROUNDED BY NAMED ABSENT REGIONS



Sequence authority. Bind the attempt. Earn settlement.

Make every transition inspectable, every authority singular, every loop finite, and every gap explicit.

THE FINAL FOLD LOCKS INTO FOUR WORDS



ONE PATH · FIVE NAMED LINKS



FINAL REFUSAL

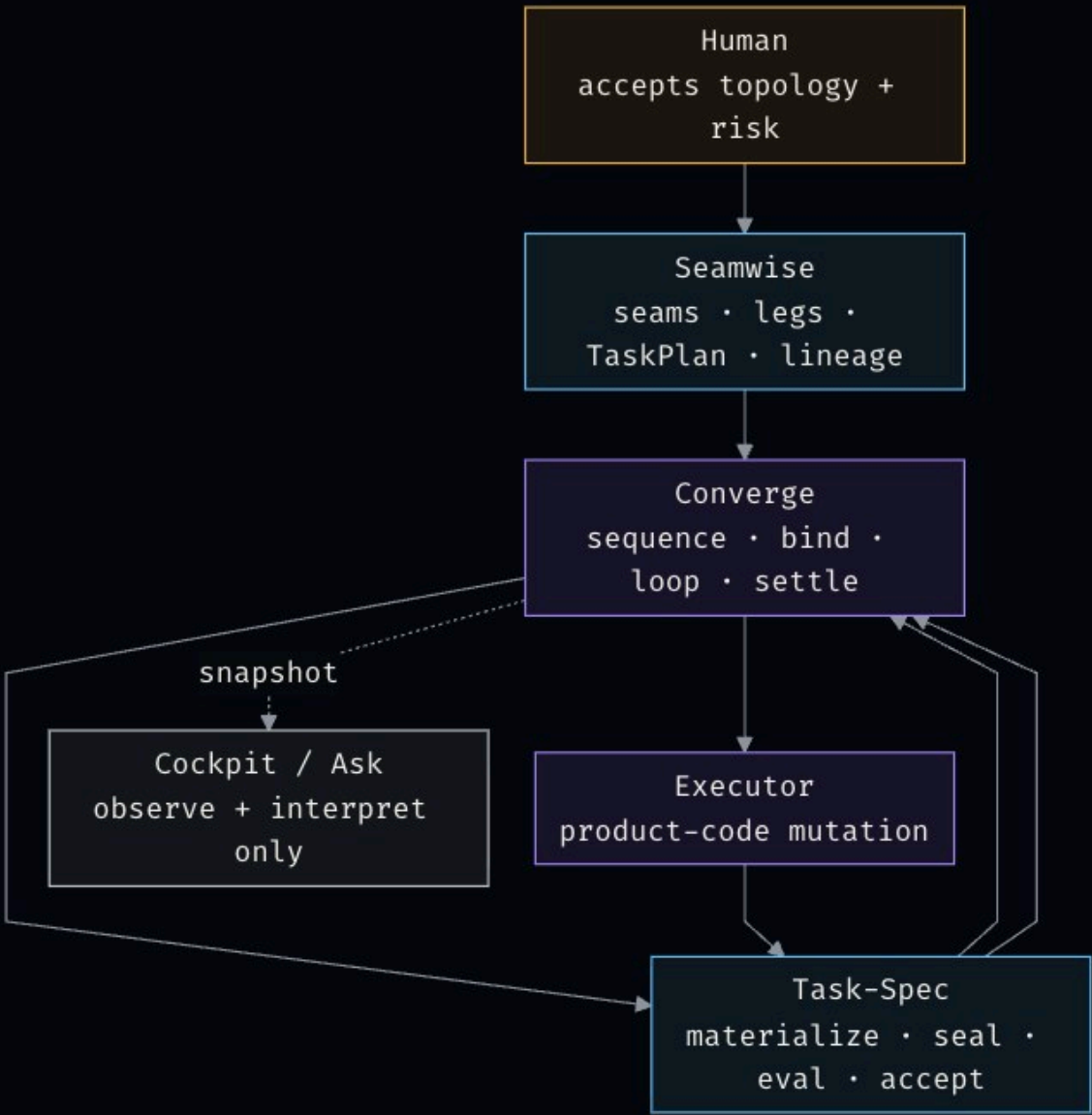
Converge can prove conformity to an encoded contract. **It cannot prove the original business decision was wise.** That judgment stays where it always lived – with the human above the barrier.

“The referee never scores its own goal.”

One authority chain, drawn seven ways.

Diagram A renders here as canonical Mermaid; B-G live on the pages that need them – **rebuilt as designed components** wherever design communicates better.

A · AUTHORITY WITHOUT DUPLICATION – RENDERED HERE



B-G · INDEXED TO THEIR PAGES

B	The descent with bypasses <i>nine passes, two bypass rails, one human barrier</i>	DESIGNED · P14
C	Compose state machine <i>five named states, one safe next action</i>	MERMAID · P26
D	Binding fences <i>task write scope vs repository gate</i>	DESIGNED · P38
E	Loop kernel <i>checkpoint → attempt → tiers → settlement</i>	MERMAID · P48
F	Receipt chain <i>seven hash-linked records, one settlement</i>	MERMAID · P54
G	Observation boundary <i>snapshot out – no mutation back</i>	MERMAID · P61
BUILD RULE · §6 Reference compositions are rebuilt with the deck's visual components ; raw Mermaid is never pasted into a slide when a designed diagram communicates the same structure better.		

Every system owns something — and explicitly does not own the rest.

Five systems, six evidence objects — the ownership cells are contractual; **read the right-hand columns as refusal lists**.

PRODUCT AUTHORITY · OWNS VS DOES NOT OWN

SYSTEM	OWNS	EXPLICITLY DOES NOT OWN
WorkHelm	everyday RPI, nuances, backlog	engine internals, skipped HMAC, factory settlement
Seamwise	evidence-backed seams, swimlanes, legs, TaskPlan, lineage	Task-Spec Markdown, dispatch, acceptance
Task-Spec	materialization, seal, handoff, eval, acceptance, TaskMesh	seam cutting, Converge settlement, user-branch merge
Converge	factory sequence, compose, bind, bounded loop, settlement	everyday coordination, topology reinterpretation, self-acceptance
Cockpit / Ask	observation and bounded interpretation	create, approve, register, bind, run, transition, settle

EVIDENCE OBJECTS · PROVES VS DOES NOT PROVE

OBJECT	PROVES	DOES NOT PROVE
Review record	named acceptance of reviewed topology digest	task authorization
Composition receipt	exact engines/source/plan/task digests materialized	dispatch or acceptance
HMAC authorization	one TaskRevision sealed under repository key	identity, secrecy, isolation, wisdom
Runtime contract	revision mapped to explicit enforcement policy	every adapter prevents violations
Execution receipt	named attempt/eval/path evidence recorded	independent acceptance
AcceptanceRecord	Task-Spec accepted exact handoff/revision	production health or universal correctness

Nine passes close on nine named gates. No gate, no descent.

Each pass turns one input into one falsifiable output; the closing gate is the **only** way out – Capture stays optional, Register stays opt-in.

PASS CONTRACTS • INPUT → OUTPUT → CLOSING GATE

PASS	INPUT	OUTPUT	CLOSING GATE
0 · Capture OPTIONAL	raw idea	BRD or no-go	CHECK_BRD=PASS
1 · Intent	signed BRD	falsifiable tech-spec	CHECK_TECH_SPEC=PASS
2 · Structure	tech-spec + repository	ADRs + glossary	CHECK_ADR=OK
3 · Decompose	requirements + terrain	swimlane/leg tree	CHECK_PLAN=OK
4 · Consensus	plans	hardened plans + decided objections	CHECK_CONSENSUS=OK
5 · Tasking	accepted TaskPlan/legs	HMAC-sealed atomic DAG	TIER=1
6 · Register OPT-IN	signed DAG	1:1 tracker projection	CHECK_REGISTER=OK
7 · Bind	one sealed revision	profile + guards + brief	CHECK_RUNTIME_CONTRACT=PASS
8 · Loop	one assigned issue	named landing + evidence	TASK_LOOP=*, then ACCEPTED=1

COMPOSE VS TASK-SPEC VS SETTLEMENT • WHO DOES WHAT

EVENT	CONVERGE	SEAMWISE	TASK-SPEC	HUMAN
Prepare	sequence	produce delivery plan	absent	inspect later
Review	record cross-engine step	record named topology acceptance	absent	decide
Preview	coordinate	compile TaskPlan	validate/preview	inspect
Materialize	receipt	provide lineage	write unsigned leaves	no dispatch yet
Authorize	route only	absent	HMAC seal	key-holder decision
Settle	bind/loop/receipt	absent	eval + independent accept	review final PR/route

Eight ways to stop — each with its exit code and its evidence.

Failure is operational: every landing names what is preserved, and every corridor of truth names **what may be claimed from it**.

TERMINAL STATES • MEANING • EXIT • WORK PRESERVED

STATE	MEANING	EXIT	WORK PRESERVED
SETTLED	accepted; external publication allowed	0	task branch + receipts
LOCAL_SETTLED	accepted; external writes denied	0	local task branch + receipts
NO_OP	already green at entry	0	existing state
BLOCKED	human/upstream/tier-2 decision needed	1	handoff; inspectable worktree when applicable
STALLED	stagnation breaker fired	1	handoff + worktree
EXHAUSTED	signed budget reached	1	handoff + worktree
CANCELLED	external stop signal	1	written checkpoint/handoff
ERROR	coordinator cannot continue safely	nonzero	only trustworthy evidence retained

RELEASE / CURRENT / CHECKOUT TRUTH • WHAT MAY BE CLAIMED

CORRIDOR	REVISION	CLI FORMS	WHAT MAY BE CLAIMED
Immutable release	v0.2.0 / 3de9f0b	release-owned surface	frozen release workflow/assets only
Committed main	58b1ddb	57	current committed code/docs/tests
Audited working tree	58b1ddb + 16 modified + 1 untracked	60	local checkout behavior, explicitly uncommitted
Future	no immutable revision	n/a	Manager/fleet and other named non-promises only as roadmap

Strong local proof, carried next to its own limits.

v0.2.0's proof is real and bounded – each verified strength travels paired with **the limit it has not crossed**.

STRONG LOCAL PROOF		EXPLICIT REMAINING LIMIT
composed single-task path	↔	no Manager/fleet scheduling
HMAC revision authorization	↔	shared-key identity/secrecy not proven
repository gate + postflight	↔	prevent class varies by adapter
optional different-family judge	↔	holdout not filesystem-separated
local settlement + independent acceptance	↔	production health not proven
Cockpit snapshot/Ask boundaries	↔	no operational write control by design
subsystem and extended gates	↔	aggregate make check red at layout

PAIRING RULE · §7

Never quote the left column without carrying the right – **the limit is part of the proof**.

Five commands compose. Six commands authorize, bind, and run.

Committed 0.2.0 surface end to end – Converge sequences, **Task-Spec seals**, and the loop receives **one named issue**.

THE COMPOSED PATH · ENGINES PINNED

composed path – cvg 0.2.0

BASH

```
export CVG_TASKSPEC_BIN=/absolute/path/to/task-spec/bin/taskspec
export CVG_SEAMWISE_BIN=/absolute/path/to/seamwise/bin/seamwise

cvg compose prepare --source recipe.yaml
cvg compose review --reviewer owner --reason "Topology and rollback accepted."
cvg compose preview
cvg compose materialize
cvg compose status
```

AUTHORIZE → BIND → RUN · STEEL THREAD

authorize · bind · run – T-20260815-health-status

BASH

```
taskspec gate --stamp cvg/tasks/T-20260815-health-status.md
cvg bind --task cvg/tasks/T-20260815-health-status.md
cvg bind --check --task cvg/tasks/T-20260815-health-status.md
git add cvg/tasks cvg/execution
git commit -m "authorize and bind health status task"
cvg loop --issue T-20260815-health-status --agent codex
```

AUTHORITY NOTE

cvg compose status is read-only – it re-hashes evidence and emits exactly one NEXT= action. Only taskspec gate --stamp seals a leaf; **composition never authorizes dispatch**.

Every call answers in one envelope. Every composition leaves one receipt.

`ConvergeCLIResult/v1` governs every CLI form; `ConvergeCompositionReceipt/v1` records materialization **without granting dispatch**.

MACHINE RESULT SHAPE · UNIVERSAL CLI ENVELOPE

ConvergeCLIResult/v1 - schema_version 1 · cvg 0.2.0 JSON

```
{
  "contract": "ConvergeCLIResult/v1",
  "ok": true,
  "token": "COMPOSE=PREVIEW_READY",
  "verdict": "PREVIEW_READY",
  "exit_code": 0,
  "changed": false,
  "dry_run": false,
  "data": {},
  "error": null,
  "meta": {"schema_version": 1, "cvg_version": "0.2.0"}
}
```

COMPOSITION RECEIPT BOUNDARY · NON-AUTHORIZING BY CONTRACT

ConvergeCompositionReceipt/v1 - dispatch not granted JSON

```
{
  "contract": "ConvergeCompositionReceipt/v1",
  "dispatch_authorized": false,
  "source_commit": "<immutable-source-commit>",
  "task_plan_digest": "sha256:<digest>",
  "task_ids": ["T-20260815-health-status"],
  "task_digests": {"T-20260815-health-status": "sha256:<digest>"}
}
```

RECEIPT ≠ AUTHORITY

A valid composition receipt proves materialization integrity – **not permission to run, not acceptance after running**. `dispatch_authorized` is contractual, and it stays false.

Deny by default. Recover by token. Mark what is uncommitted.

The loop's safety posture in three specimens – external writes denied, recovery driven by the machine envelope, and checkout-only control **visibly marked W**.

DEFAULT-DENY LOOP POLICY

loop policy – default denyYAML

```
policy:
  external_writes: deny
capabilities:
  grants: []
budgets:
  iterations: 15
  wall_seconds: 5400
  tokens: 120000
isolation: worktree
```

RECOVERY BY TOKEN

recovery – by tokenBASH

```
cvsg compose status --json
# Branch on token, exit_code, changed, dry_run.

cvsg loop --estimate --issue T-20260815-health-status
cvsg loop --gate-only --issue T-20260815-health-status
# Never convert BLOCKED/STALLED/EXHAUSTED into success.
```

W CHECKOUT ONLY

guided chat – W · uncommittedBASH

```
# W · uncommitted at the audited checkout
cvsg next --guided --lane FULL
cvsg next pre 5 --lane FULL
cvsg next post 5
```

READING THE POLICY

grants: [] plus external_writes: deny means the loop settles locally by default; --estimate and --gate-only spend no model budget. **Failure tokens are never converted into success.**

Four presenter routes; eight sanctioned demo stops.

Pick the route for the room — then place live specimens **exactly where the evidence lands**.

PRESENTER ROUTES · § 9

ROUTE	SLIDES	TIME	USE
Executive	01–08, 22, 29–30, 37, 47, 54, 59–60, 72–76	25–30 min	category, authority, proof boundary, current truth
Method	01–36, 47, 54, 59–60, 75–76	50–60 min	nine-pass teaching without runtime internals
Engineering	all 76	95–120 min	full architecture, contracts, trust, operations, release audit
Workshop	all 76 + live specimens	150–180 min	stop after each act for repository inspection

DEMO PLACEMENT · EIGHT STOPS

- 1

After S08: run `cvg agent-context --json` to show the product surface.
- 2

After S22: show a decided objection log, then mutate a reviewed plan in a disposable fixture and rerun the barrier gate.
- 3

After S30: inspect an unsigned materialized leaf and composition receipt.
- 4

After S46: run `cvg bind --check` and show that Git remains unchanged.
- 5

After S54: run `cvg loop --estimate` rather than spending a live model budget.
- 6

After S60: show the retained `v0.2.0/live-codex` receipt chain.
- 7

After S67: open Cockpit against a disposable workspace; never point the demo at **sensitive customer evidence**.
- 8

Before S74: show the exact `LAYOUT=FAIL` output. Do not "fix live" during the presentation.

Five questions the room must answer. Ten checks every slide must pass.

Run the checks before, during, and after presenting – **the expected answers are the book's load-bearing claims.**

AUDIENCE CHECKS · §9

"Which artifact authorizes dispatch?"

Expected: the Task-Spec **HMAC seal** – not review, materialization, or a receipt.

"Who accepts the result?"

Expected: the Task-Spec **POST gate**; Converge coordinates and records settlement.

"Does Cockpit control the run?"

Expected: **no** – read-only projection and interpretation only.

"What does LOCAL_SETTLED exclude?"

Expected: **external publication and tracker mutation**, unless separately authorized.

"What is the current aggregate status?"

Expected: subsystem proof is broad; **make check is red at layout.**

PER-SLIDE ACCEPTANCE CHECKLIST · §10

- ✓ One slide, **one authority decision** or trust boundary.
- ✓ The headline is a **claim**, not a topic label.
- ✓ The visual expresses the relationship **without requiring notes**.
- ✓ Every green statement names a **source label and proof token**.
- ✓ Release, current main, working tree, and future claims are **visibly distinct**.
- ✓ No receipt is shown **granting authority it does not own**.
- ✓ No dotted observation line **looks like a write path**.
- ✓ Code fits without horizontal scroll at **1366×768**.
- ✓ Mobile order preserves **semantic sequence**.
- ✓ The transition creates **one question** answered by the next slide.

Every claim in this book traces to one of these paths.

The footer labels resolve here – the index is the book's **chain of custody**, audited at `58b1ddb` on 2026-08-26.

AUTHORITY AND OPERATING MODEL

- A** OPERATING.md – *everyday/factory split, ownership, merge rule, fail-closed hop rule*
- A** AGENTS.md – *read-order and non-negotiable entry rules*
- A** docs/concepts/authority.md – *sole-authority table and executable boundary enforcement*
- D** docs/trust/index.md – *what each mechanism proves and refuses*

PRODUCT AND MACHINE CONTRACTS

- C** VERSION, package.json, .claude-plugin/ – *package identity and install surface*
- C** contracts/cli-command-matrix.json – *command names, pass mapping, mutation flags, tokens*
- C** contracts/converge-cli-result-v1.schema.json – *universal CLI result*
- C** contracts/converge-composition-receipt-v1.schema.json – *non-authorizing composition receipt*
- C** contracts/ui/v3/workspace-snapshot.schema.json – *observer read model*

CLI, COMPOSITION, AND SEQUENCING

- I** bin/cvg – *stable referee, routing, setup, doctors, global JSON/dry-run behavior*
- I** bin/_cvg_compose.py – *prepare/review/preview/materialize/status state machine*
- I** bin/cvg-classify-lane.py – *FAST/NORMAL/FULL routing*
- I** bin/cvg-snapshot.py – *canonical read-only workspace projection*
- I** skills/evidence-to-next-pass/ – *evidence-derived sequence and pass prompts*

PASS IMPLEMENTATION

- I** skills/idea-to-brd/ – *optional Capture*
- I** skills/brd-docs-to-tech-req/ – *Intent*
- I** skills/tech-req-to-adrs/ – *Structure*
- I** skills/reqs-to-swimlane-plans/ – *Decompose*
- I** skills/sketch-plans-adversarial-review/ – *Consensus and barrier*
- I** external Task-Spec 3.8.x – *Tasking, HMAC seal, handoff, eval, acceptance*
- I** skills/task-specs-to-issues/ – *optional Register*
- I** skills/task-to-runtime-contract/ – *Bind, repository gate, adapters, tier 2*
- I** skills/task-loop/ – *finite loop and settlement*

OPERATIONS AND OBSERVATION

- D** docs/guides/descent.md – *pass map, optional/opt-in routes, terminal states*
- D** docs/guides/bind-and-loop.md – *authorize/bind/loop/verify/accept route and holdout caveat*
- D** docs/guides/recovery.md – *one safe next action and resume limitation*
- I** apps/cockpit/ – *read-only UI, transport, artifact bridge, ACP Ask path*
- I** install.sh, templates/ – *consumer projection and workspace policy*

DIRECT PROOF

- T** Makefile – *local and release gate composition*
- T** tests/test-compose.sh – *review/materialization/receipt state machine*
- T** tests/test-clean-room-install-e2e.sh – *install-to-acceptance steel thread*
- T** tests/test-loop-kernel.sh – *budgets, isolation, refutation, external writes, terminal states*
- T** skills/task-to-runtime-contract/tests/run-tests.sh – *binding and policy boundary*
- T** skills/task-specs-to-issues/tests/test-register.sh – *tracker projection parity*
- T** tests/test-cvg-json-envelope.sh, tests/test-cvg-json-matrix.py – *machine surface*
- T** tests/test-repo-layout.sh – *current aggregate blocker*
- T** Cockpit server/Vitest suites – *read-only bridge, Ask, transport, UI*

RELEASE AND RETAINED EVIDENCE

- R** tag v0.2.0 at 3de9f0b5f83f1bb62475308317c58e53f851b0db
- R** .github/workflows/release.yml – *immutable tag verification and publication route*
- E** evidence/releases/v0.2.0/live-codex/ – *retained valid composed execution*
- E** invalidated alpha corridors – *dirty-engine and pipe-hang attempts are not silently upgraded*
- D** README.md release truth – *reported hosted run; refresh live before a current hosted claim*

Sixteen rules this book held itself to. Any page breaking one is a defect.

The editorial contract, condensed from §12 – **claim discipline is part of the product.**

RULE 01

Name it **factory coordinator, assurance layer, or referee** — never the autonomous reasoning engine.

RULE 02

Say "**nine passes, two phases, one barrier**"; Capture optional and Register opt-in whenever the full spine appears.

RULE 03

Keep topology acceptance, task authorization, runtime binding, execution, acceptance, and observation **visually separate**.

RULE 04

`reviewed`, `materialized`, `green`, `settled`, `accepted` are **never synonyms**.

RULE 05

`LOCAL_SETTLED` when external writes are denied; `SETTLED` only for **explicitly permitted publication**.

RULE 06

Task-Spec `3.8.x` compatibility stays explicit — **no standalone 3.9 behavior** in Converge 0.2.0.

RULE 07

Guided chat and 60-form CLI content stay **W • CHECKOUT ONLY** until committed and re-proven.

RULE 08

Hosted/release statements are **reported or immutable**; local reruns do not prove remote publication.

RULE 09

The layout failure lives in the **main narrative**, not a footnote.

RULE 10

HMAC is **tamper evidence** under a shared key — not identity, not isolation.

RULE 11

Tier-2 holdout secrecy is **incomplete** until the worker cannot read judge-only criteria.

RULE 12

Adapter enforcement is `prevent` or `detect` — **exactly as recorded**.

RULE 13

Cockpit, Ask, tracker, and `cvg next` **never write canonical truth**.

RULE 14

Converge never **merges the user branch**, chooses portfolio work, or silently widens authority.

RULE 15

Failure tokens and exit semantics are preserved — an error or exhausted budget is **never success**.

RULE 16

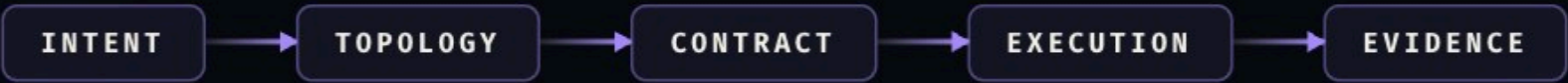
End with the honest promise: Converge proves **conformity to the encoded contract**, not the wisdom of the original decision.



Settlement Fold

PROCESS IDENTITY · THE FOLD THAT SETTLES

Sequence authority. Bind the attempt. Earn settlement.



Built from brf-converge.md — the deep-dive presentation architecture
Converge 0.2.0 · Task-Spec 3.8.x · Seamwise pairing 0.2.0
Source audited 2026-08-26 at 58b1ddb · release anchor v0.2.0 → 3de9f0b